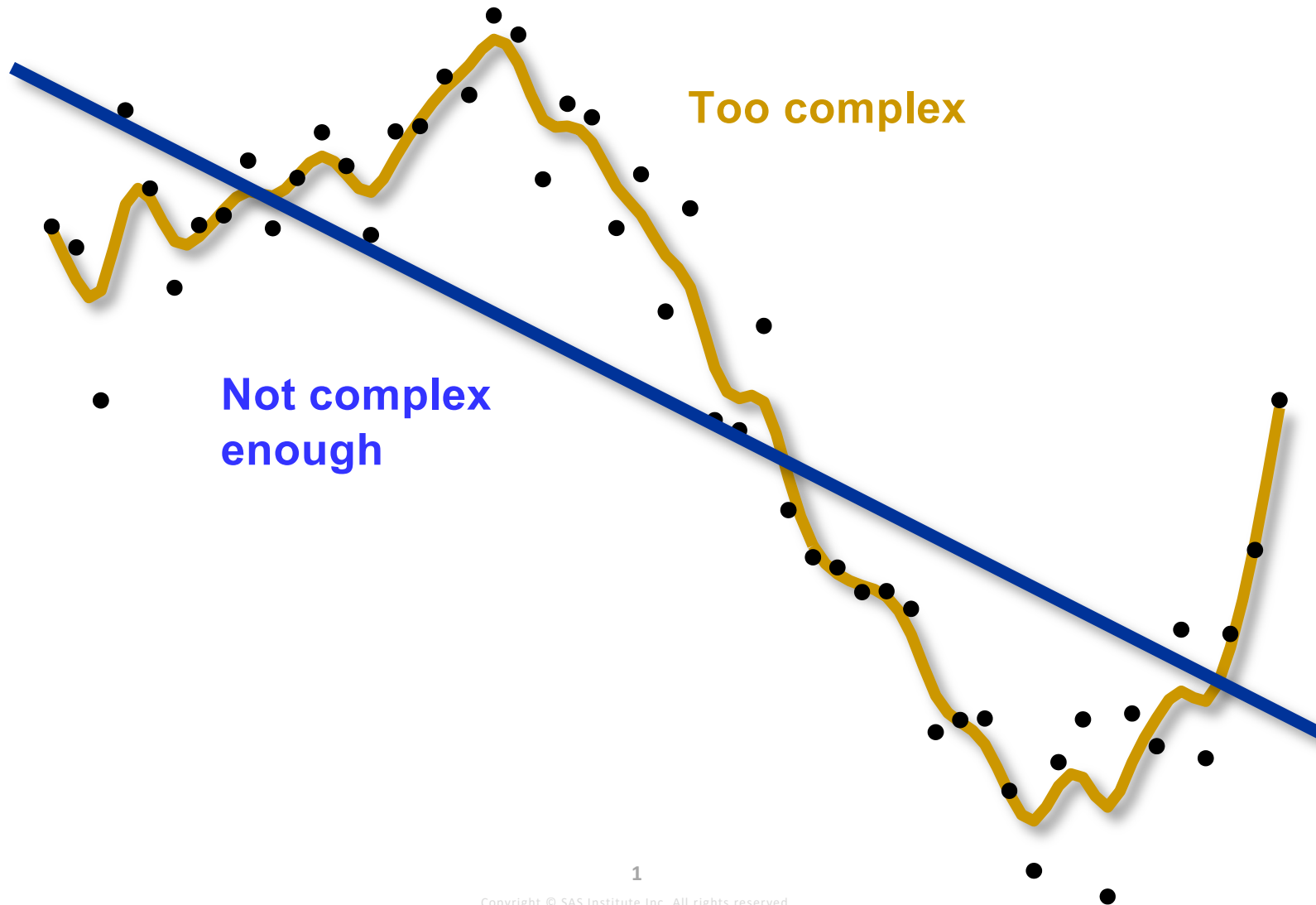


Model Complexity



Data Partitioning

Training Data

	inputs			target

Validation Data

	inputs			target

Partition available data into training and validation sets.

Data Partitioning

Training Data

	inputs			target

Validation Data

	inputs			target

Partition available data into training and validation sets.

Predictive Model Sequence

Training Data

	inputs			target

Validation Data

	inputs			target



**Create a
sequence of
models with
increasing
complexity
from the
training data.**

***Model
Complexity***

Predictive Model Sequence

Training Data

	inputs			target

Validation Data

	inputs			target



**Create a
sequence of
models with
increasing
complexity
from the
training data.**

**Model
Complexity**

Model Performance Assessment

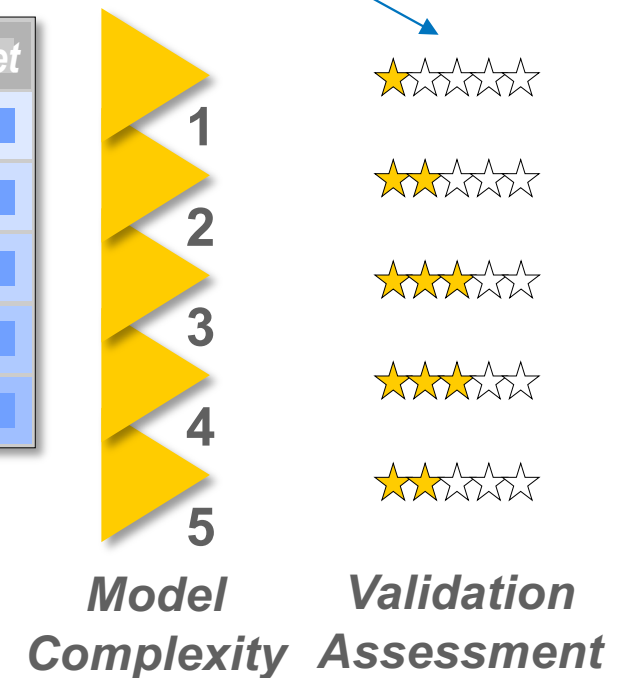
Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

**Rate model performance
using *validation* data.**



Model Performance Assessment

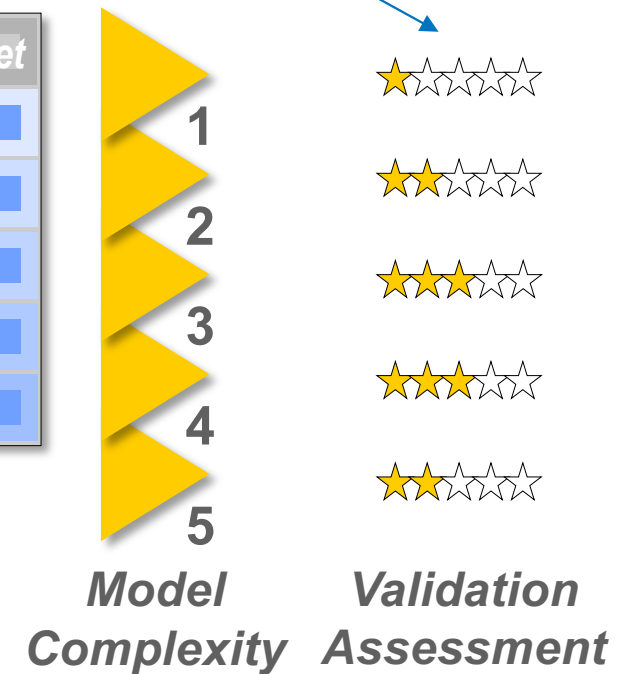
Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

**Rate model performance
using *validation* data.**



Model Selection

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

Select the simplest model with the highest *validation* assessment.



Model Complexity



Validation Assessment

Model Selection

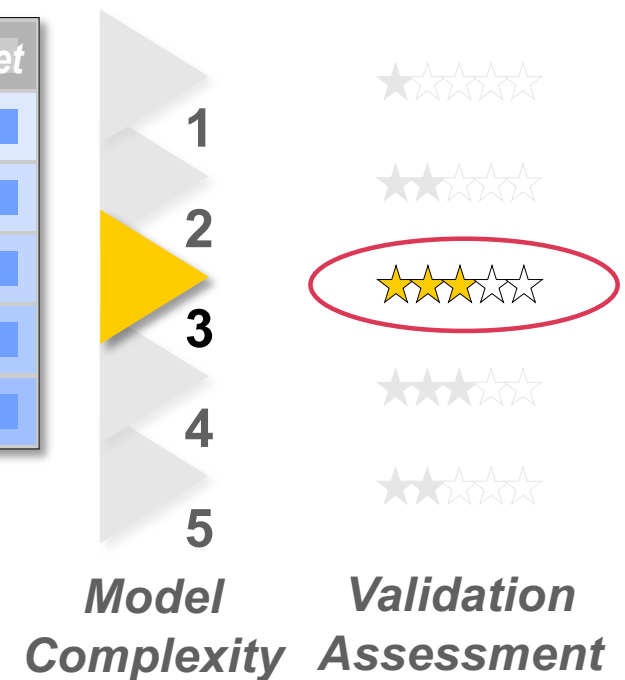
Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

Select the simplest model with the highest *validation* assessment.



Modeling Essentials: Optimize Review

▶ Predict new cases.

Decide, rank,
and estimate.

▶ Select useful inputs.

Eradicate
redundancies
and irrelevancies.

▶ **Optimize complexity.**

**Tune models with
validation data.**



Creating Training and Validation Data

This demonstration illustrates how to partition a data source into training and validation sets.

Lesson 3: Introduction to Predictive Modeling: Predictive Modeling Fundamentals and Decision Trees

3.1 Introduction

3.2 Cultivating Decision Trees

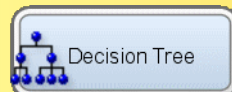
3.3 Optimizing the Complexity of Decision Trees

3.4 Understanding Additional Diagnostic Tools (Self-Study)

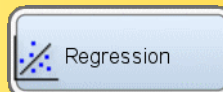
3.5 Autonomous Tree Growth Options (Self-Study)

Predictive Modeling Tools

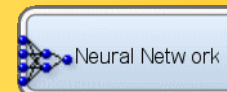
Primary



Decision Tree

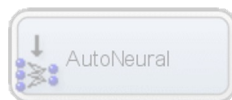


Regression



Neural Network

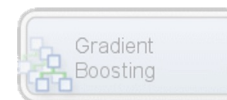
Specialty



AutoNeural



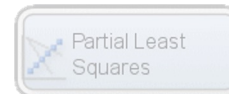
Dmine
Regression



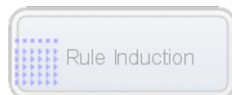
Gradient
Boosting



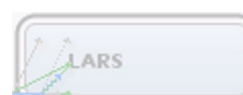
MBR



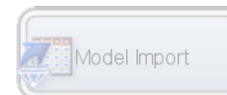
Partial Least
Squares



Rule Induction

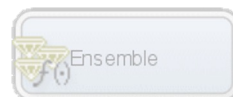


LARS



Model Import

Multiple



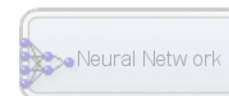
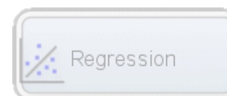
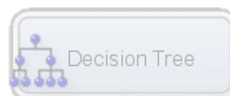
Ensemble



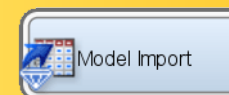
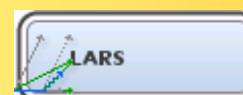
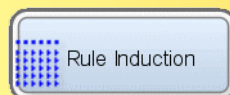
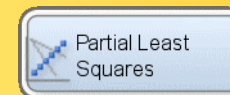
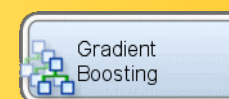
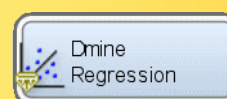
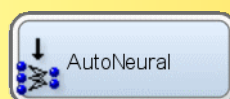
TwoStage

Predictive Modeling Tools

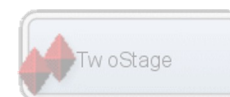
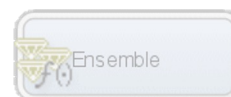
Primary



Specialty

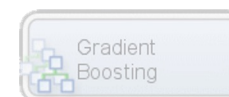
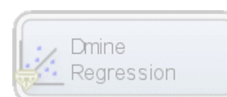
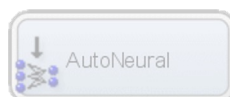
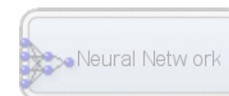
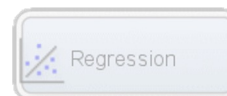
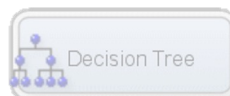


Multiple

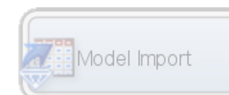
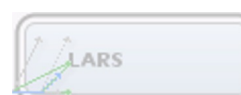
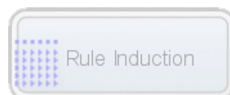


Predictive Modeling Tools

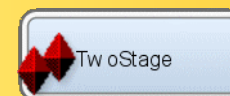
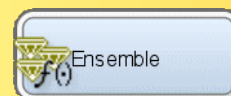
Primary



Specialty



Multiple



Model Essentials: Decision Trees

- ▶ **Predict new cases.**
- ▶ **Select useful inputs.**
- ▶ **Optimize complexity.**

Prediction rules

Split search

Pruning

Model Essentials: Decision Trees

- ▶ **Predict new cases.**
- ▶ **Select useful inputs.**
- ▶ **Optimize complexity.**

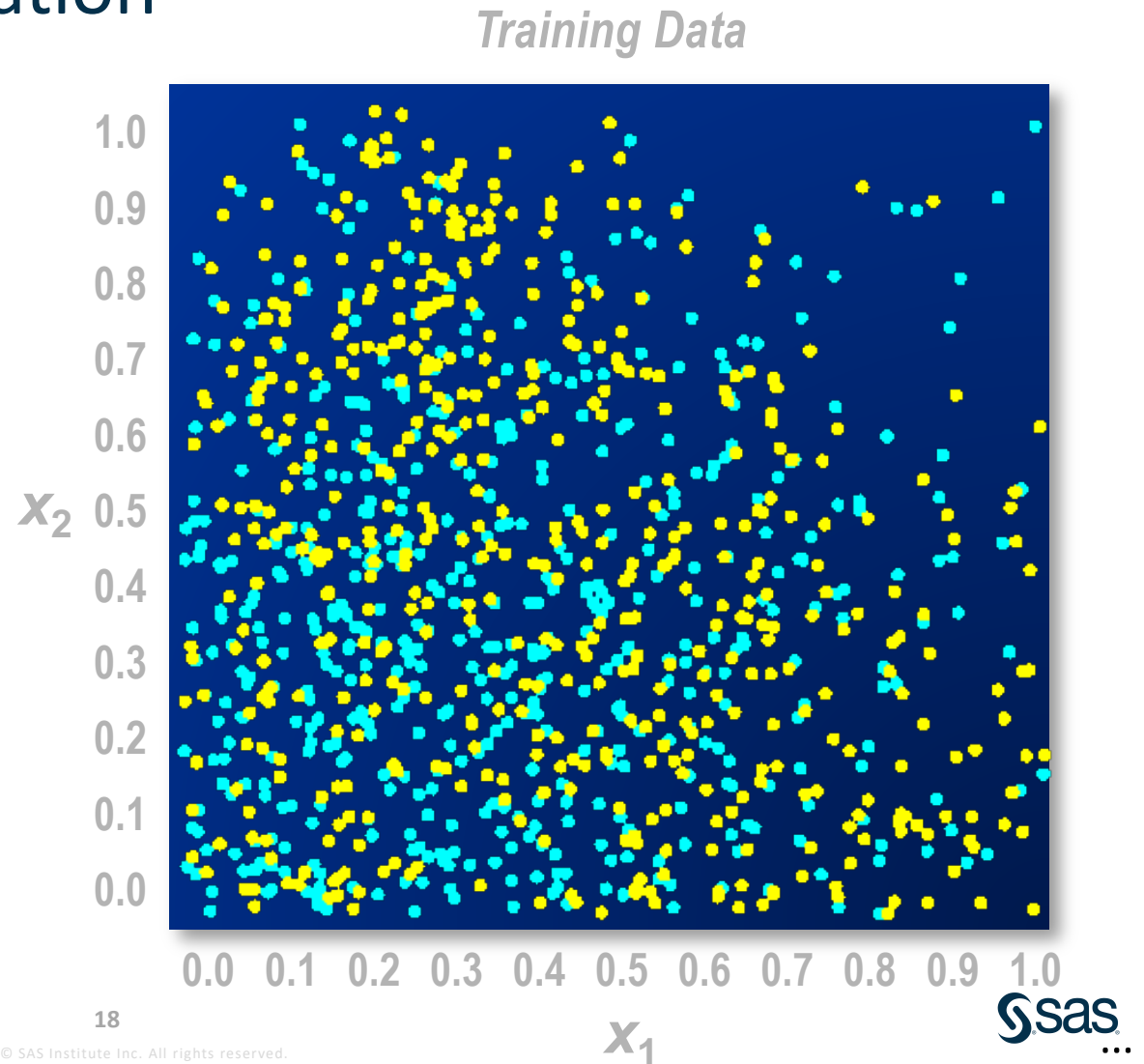
Prediction rules

Split search

Pruning

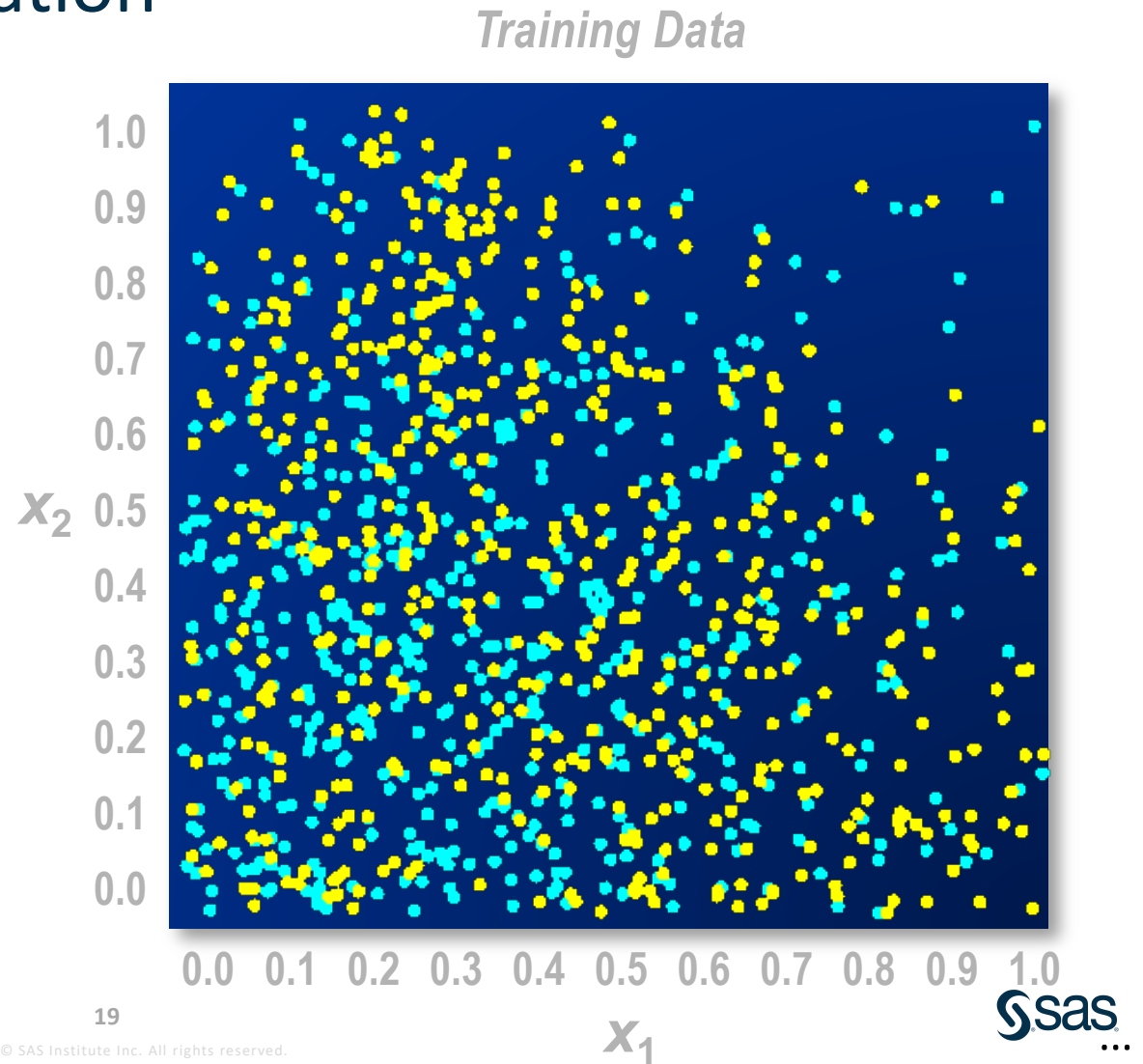
Simple Prediction Illustration

**Predict dot color
for each x_1 and x_2 .**

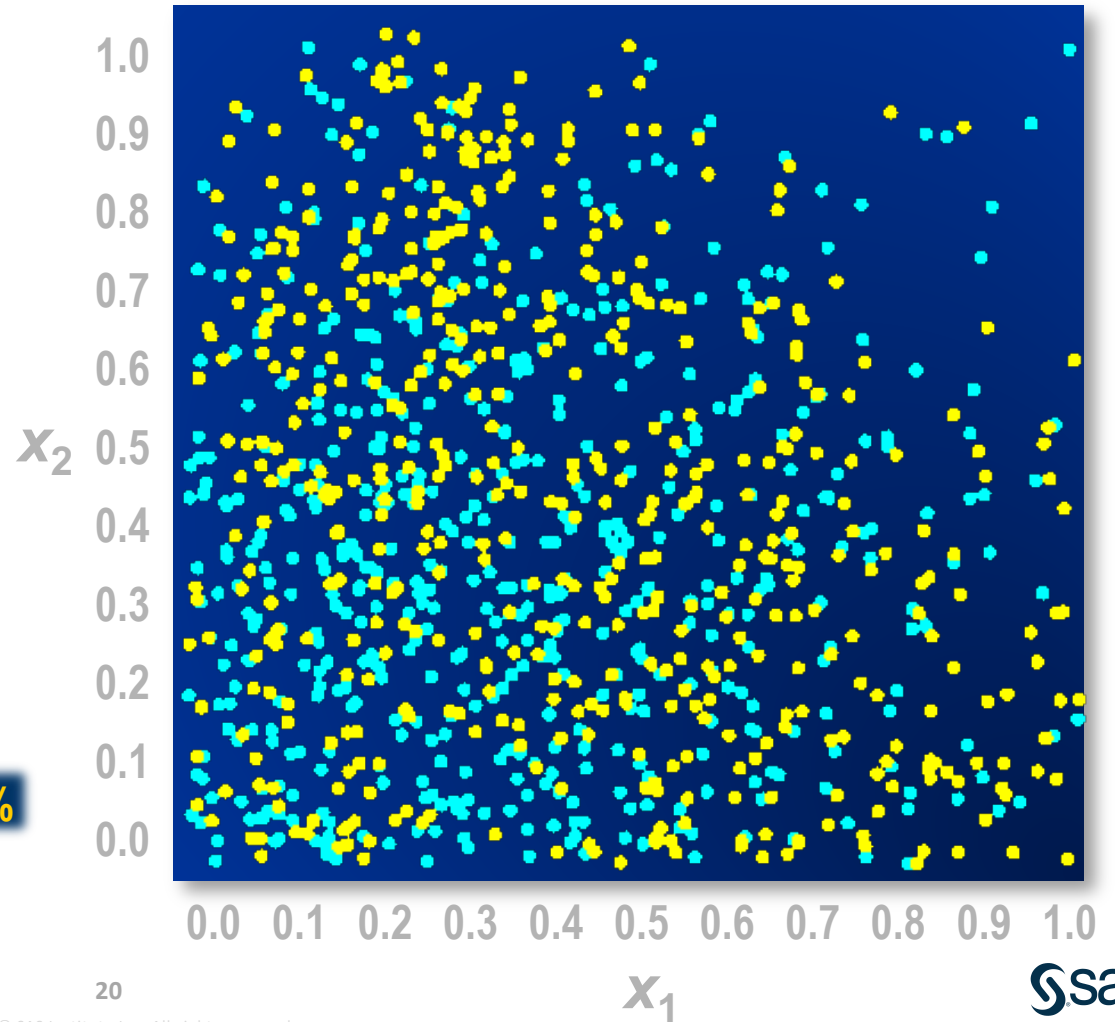
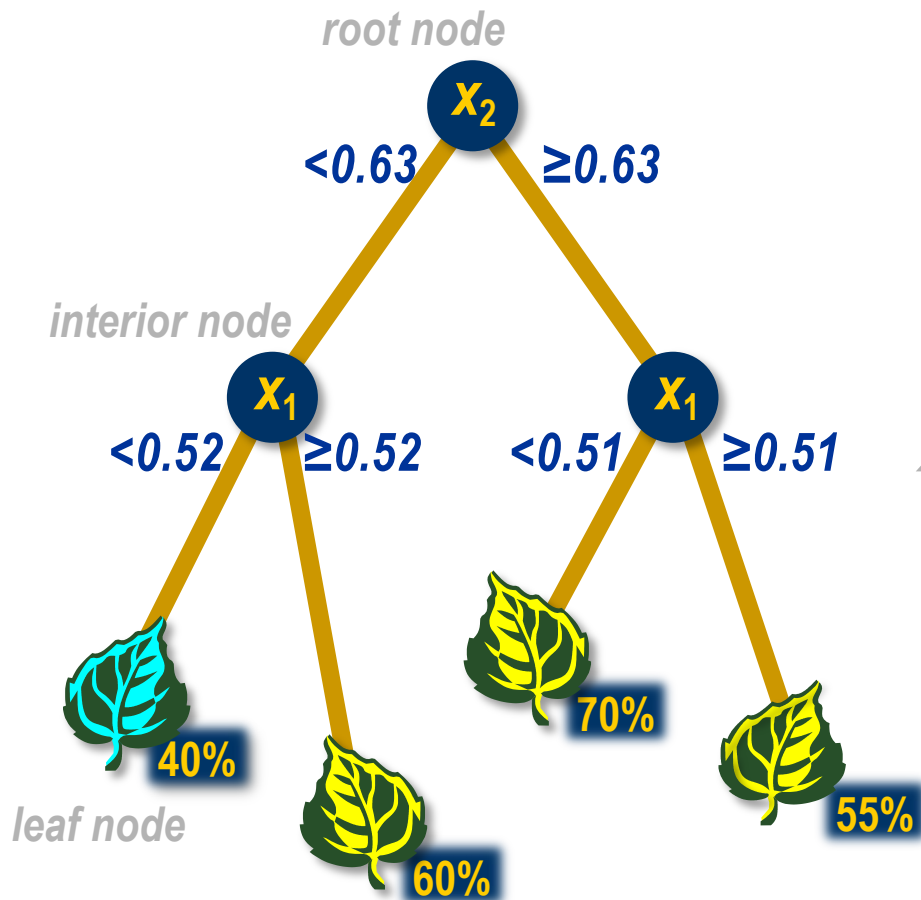


Simple Prediction Illustration

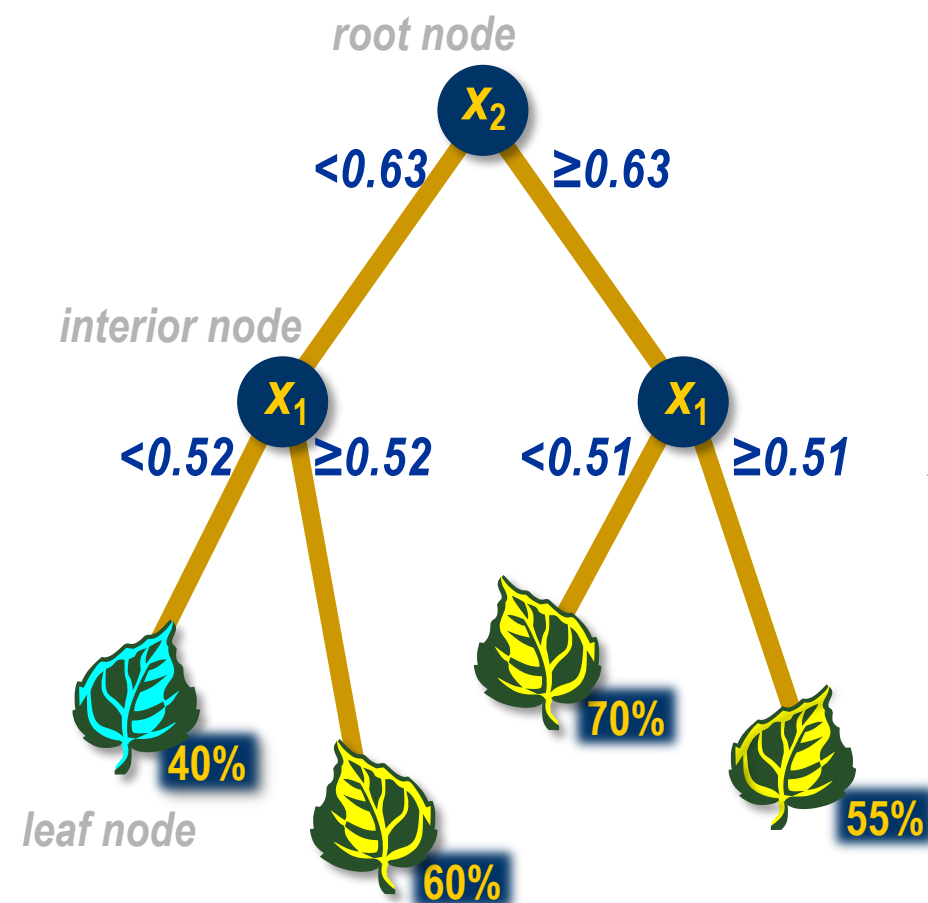
**Predict dot color
for each x_1 and x_2 .**



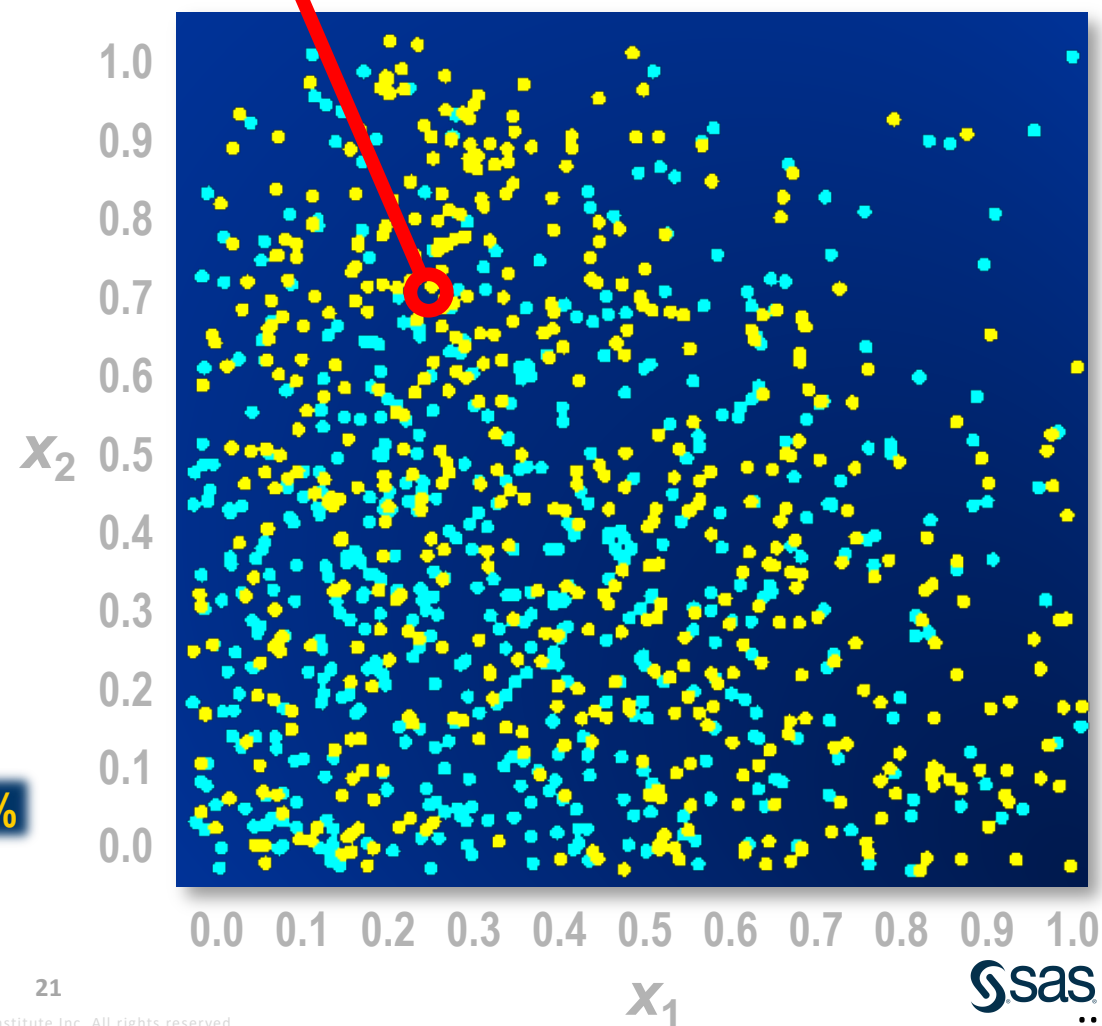
Decision Tree Prediction Rules



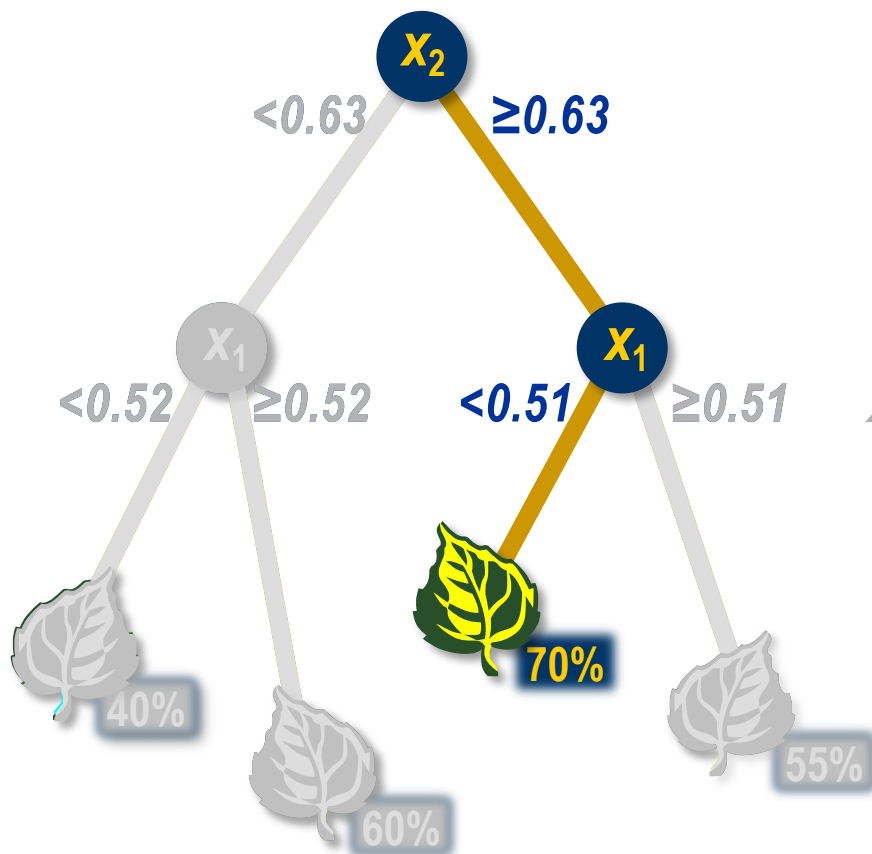
Decision Tree Prediction Rules



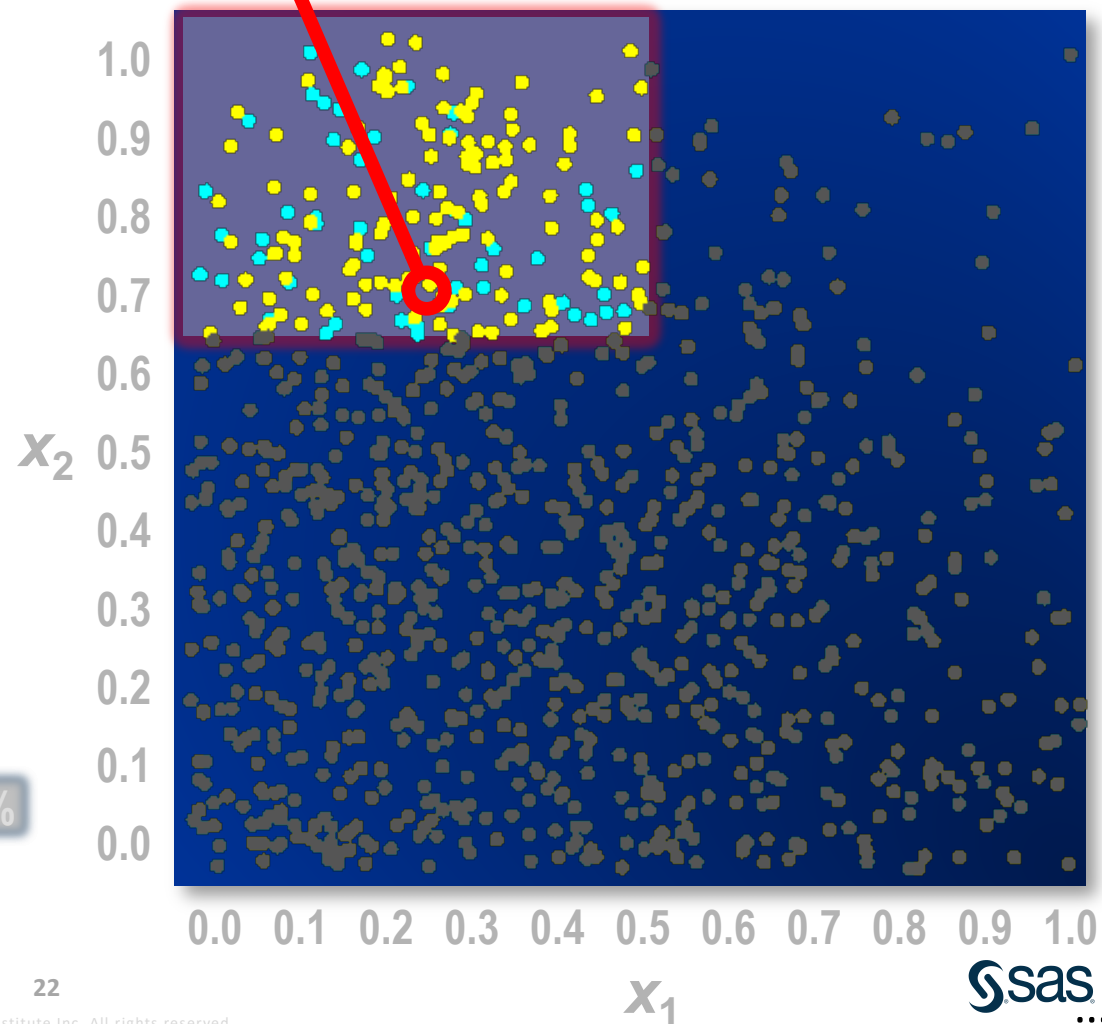
Predict:



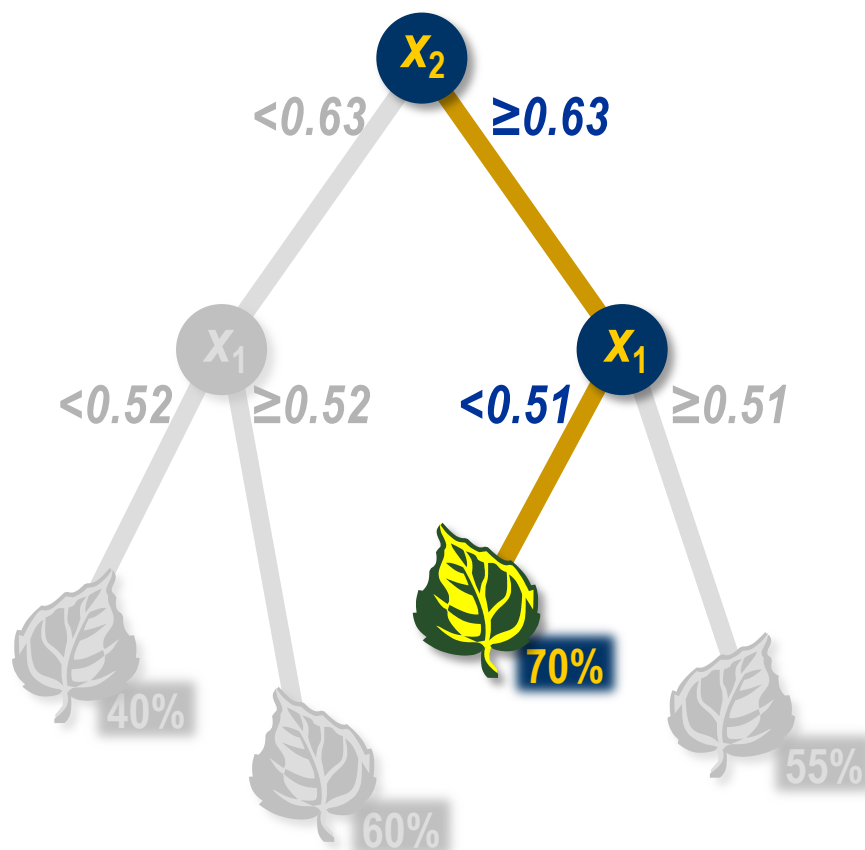
Decision Tree Prediction Rules



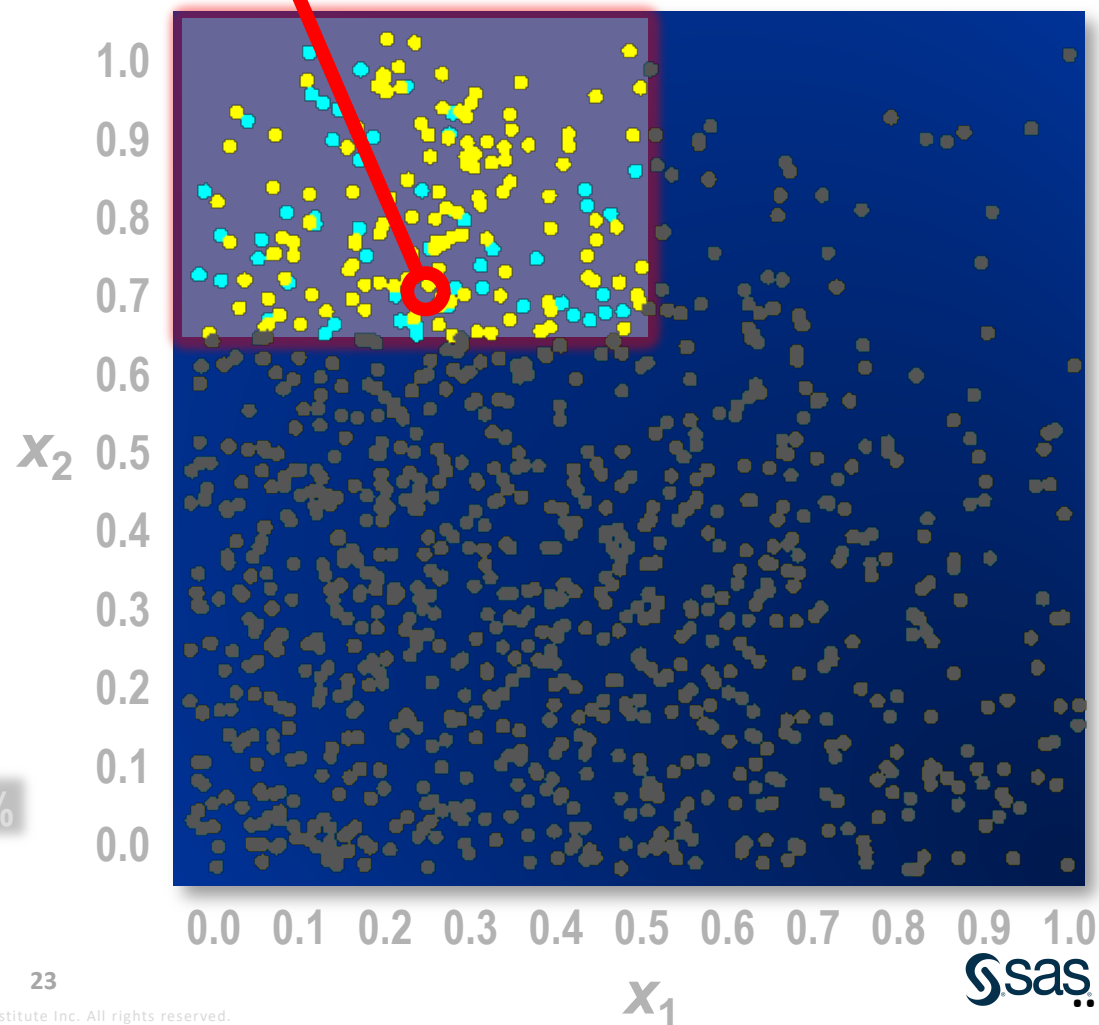
Predict: Decision = ●
Estimate = 0.70



Decision Tree Prediction Rules



Predict: Decision = ●
Estimate = 0.70



Model Essentials: Decision Trees

▶ **Predict new cases.**

Prediction rules

▶ **Select useful inputs.**

Split search

▶ **Optimize complexity.**

Pruning

Model Essentials: Decision Trees

▶ **Predict new cases.**

Prediction rules

▶ **Select useful inputs.**

Split search

▶ **Optimize complexity.**

Pruning

Model Essentials: Decision Trees

▶ Predict new cases.

Prediction rules

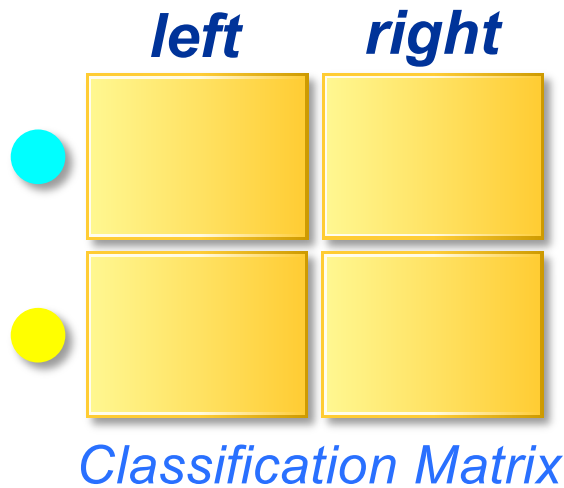
▶ **Select useful inputs.**

Split search

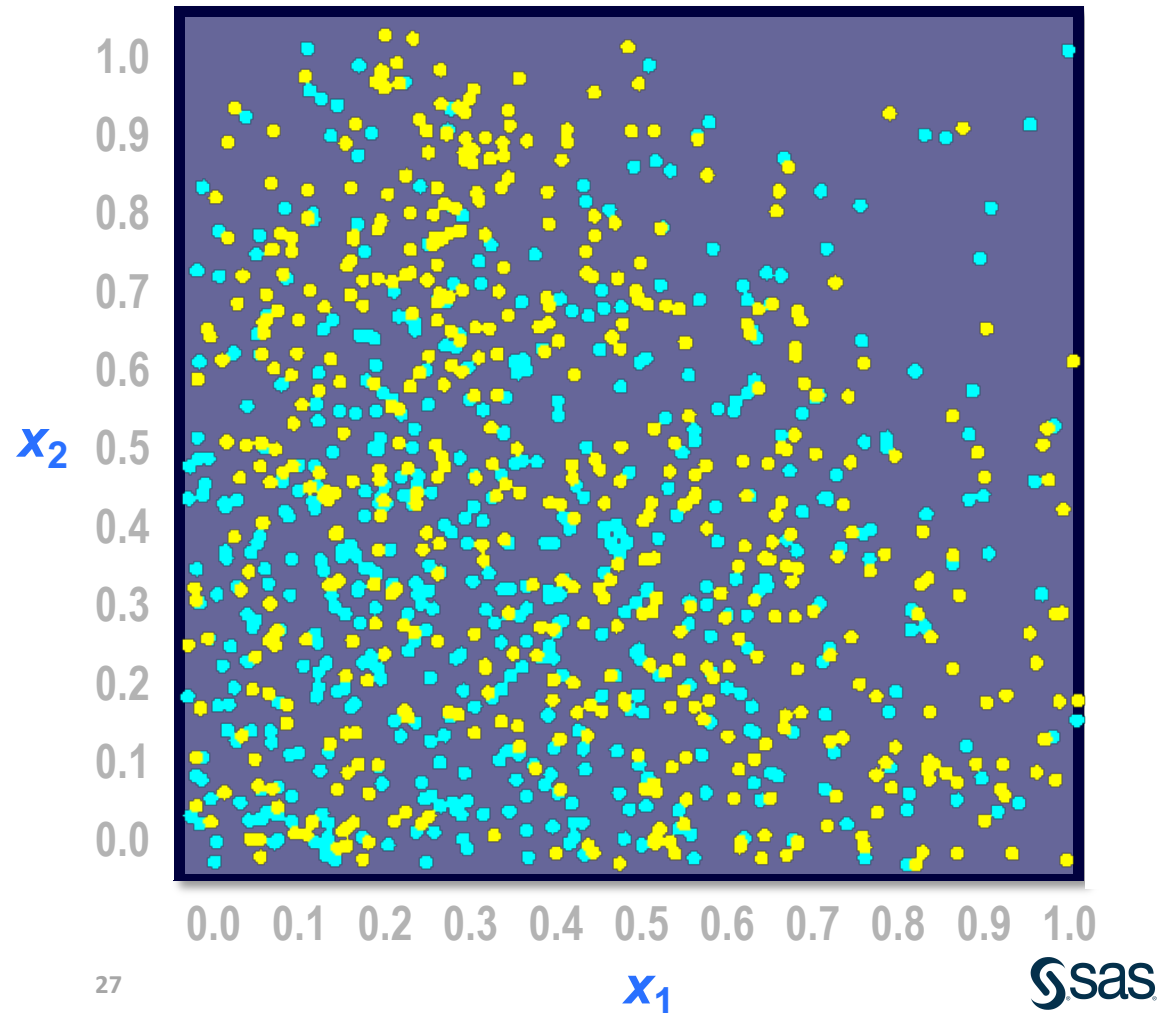
▶ Optimize complexity.

Pruning

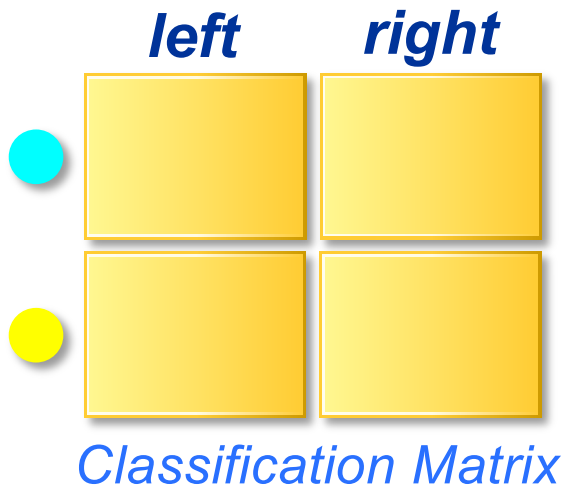
Decision Tree Split Search



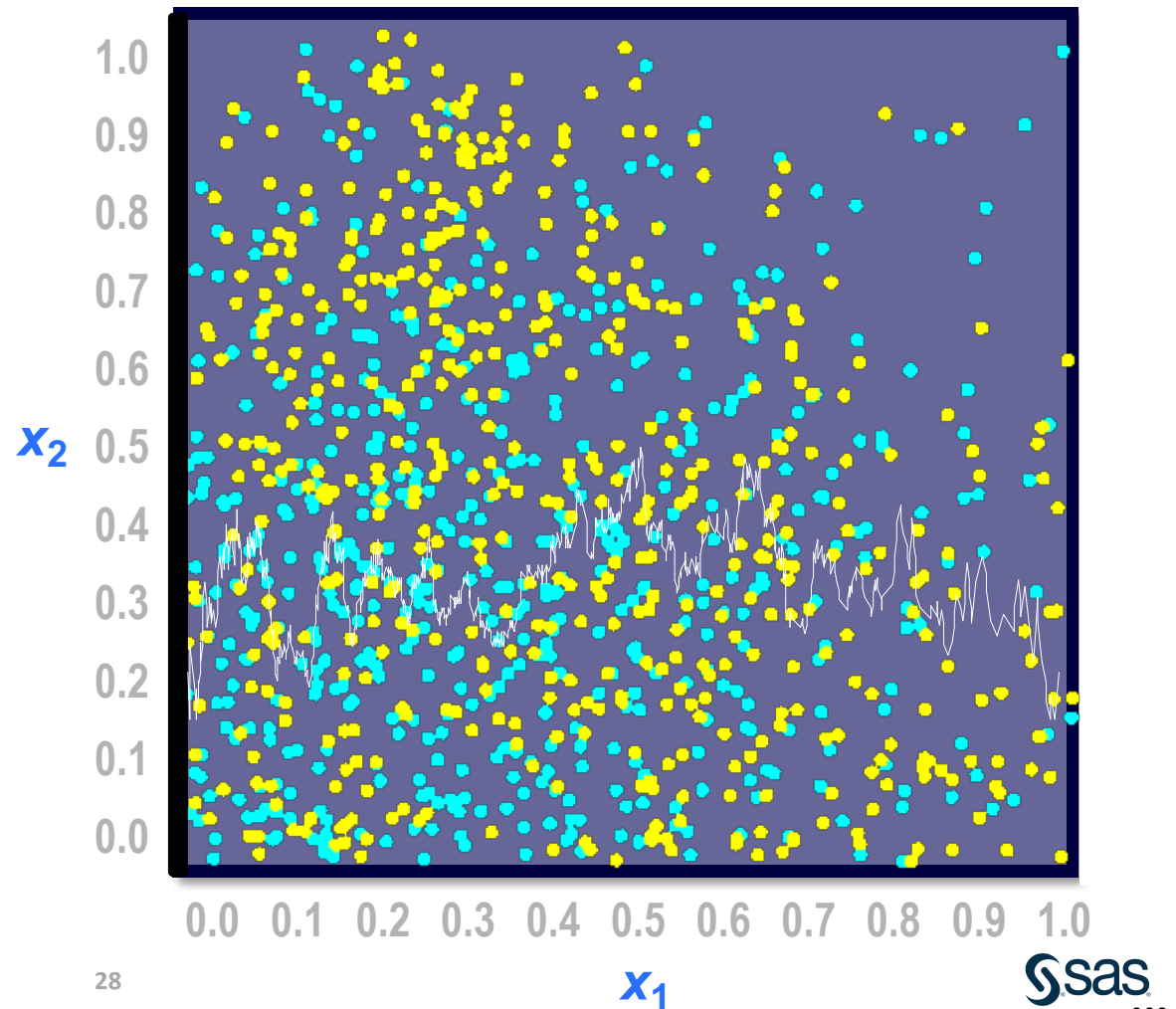
Calculate the **logworth** of every partition on input x_1 .



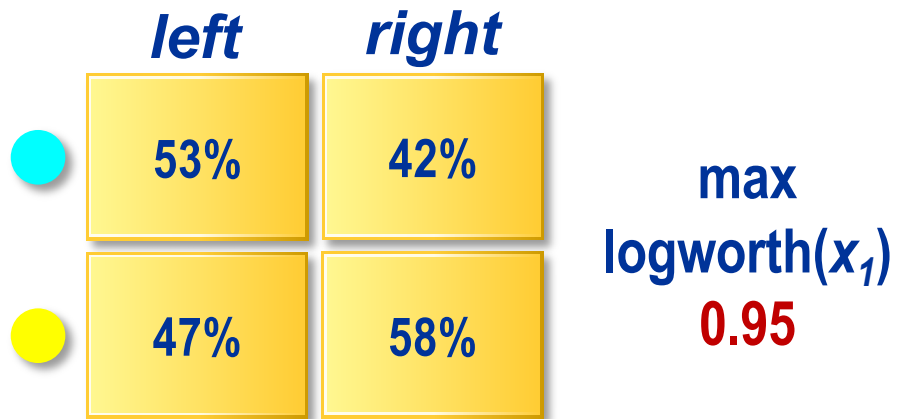
Decision Tree Split Search



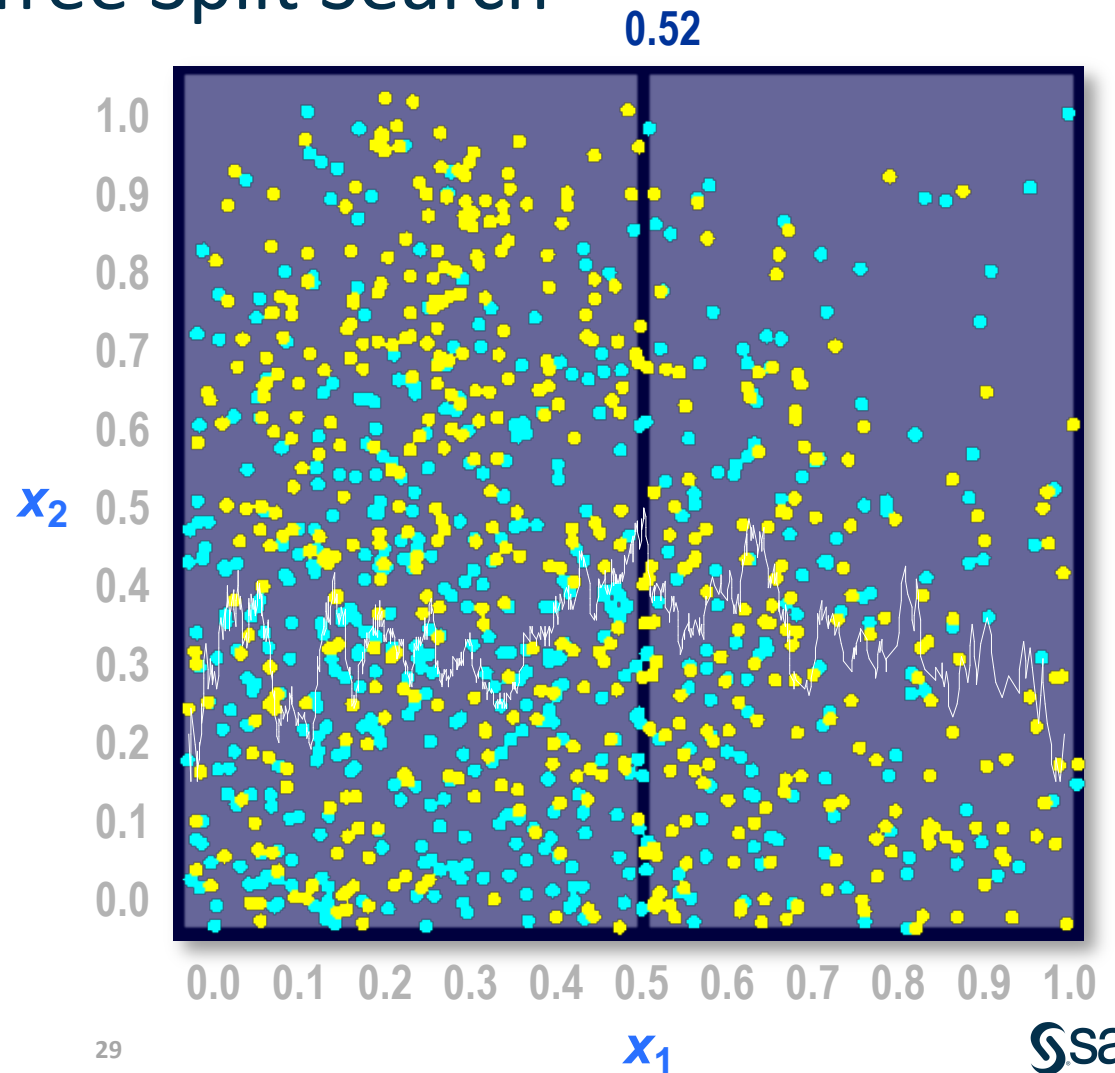
Calculate the **logworth**
of every partition on
input x_1 .



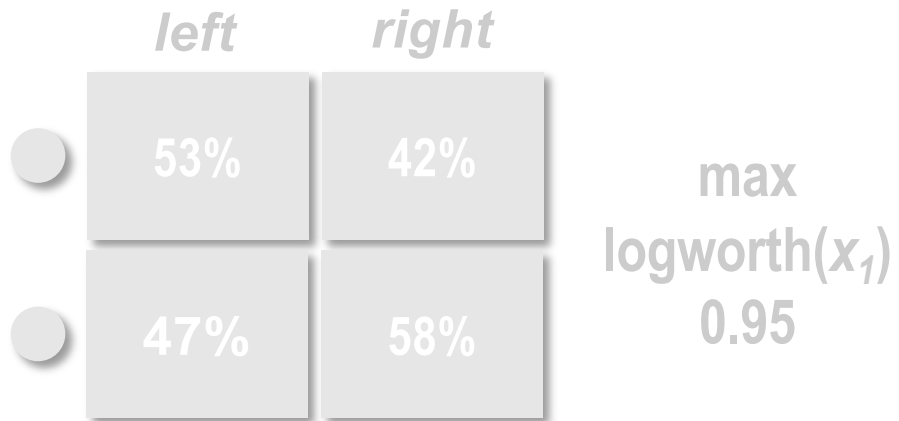
Decision Tree Split Search



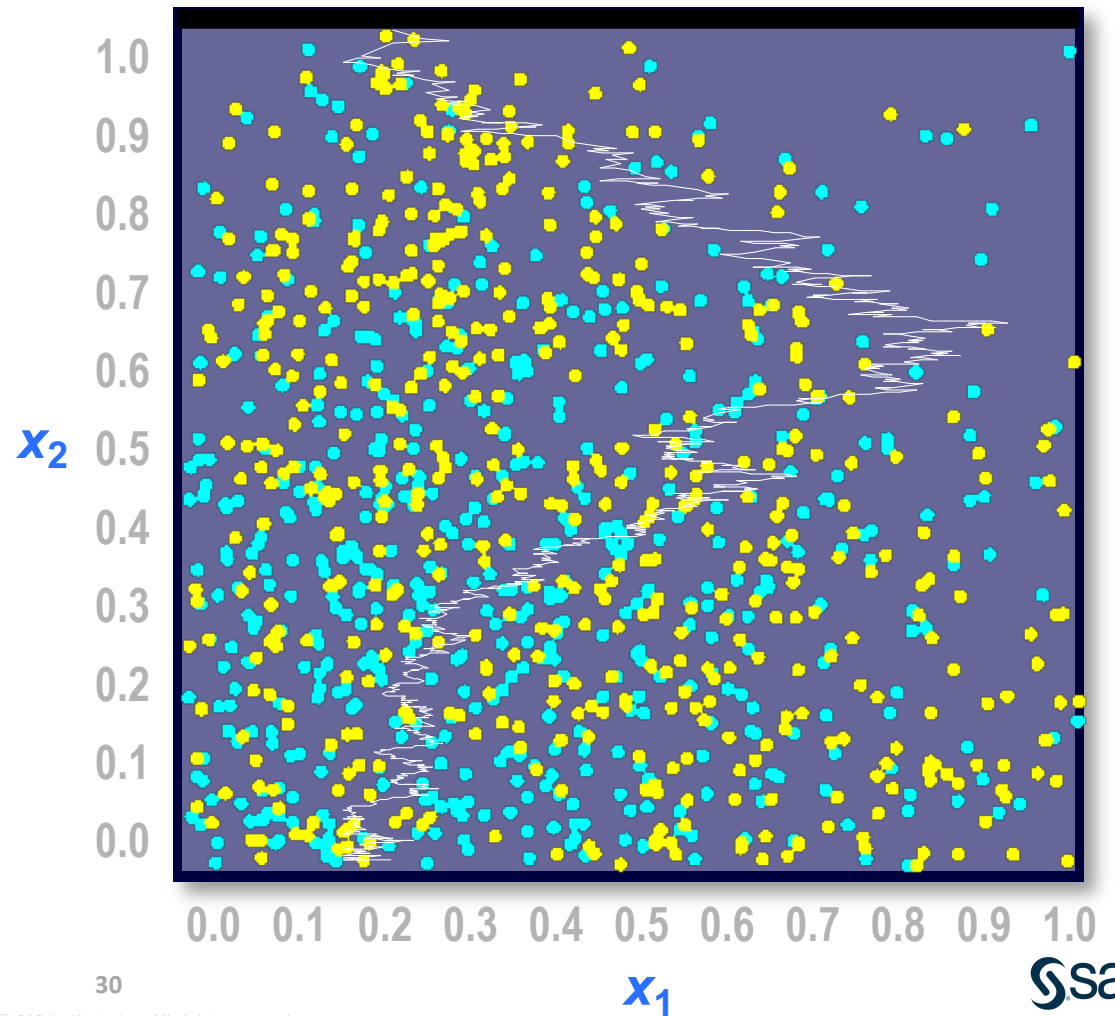
Select the partition with
the maximum *logworth*.



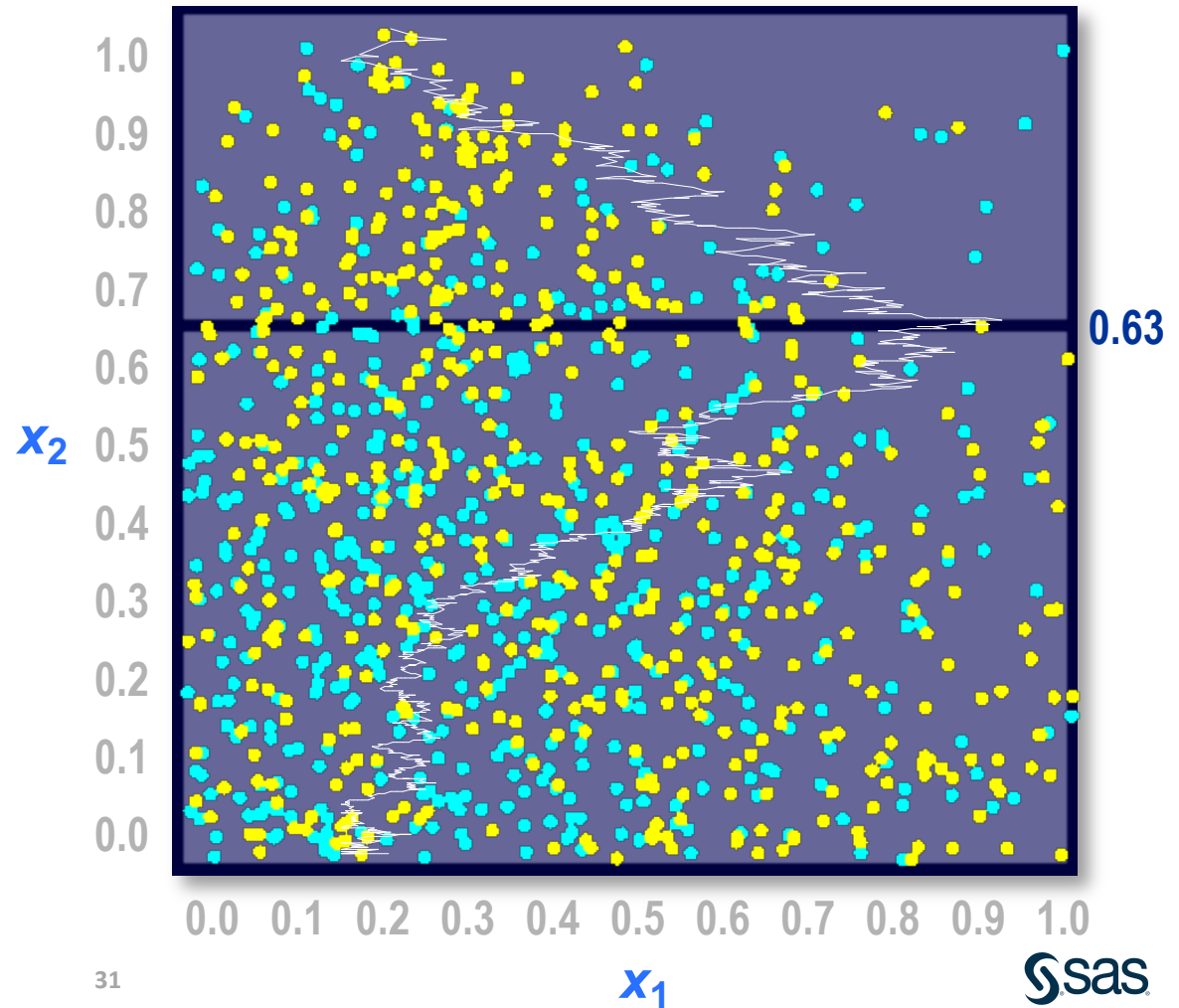
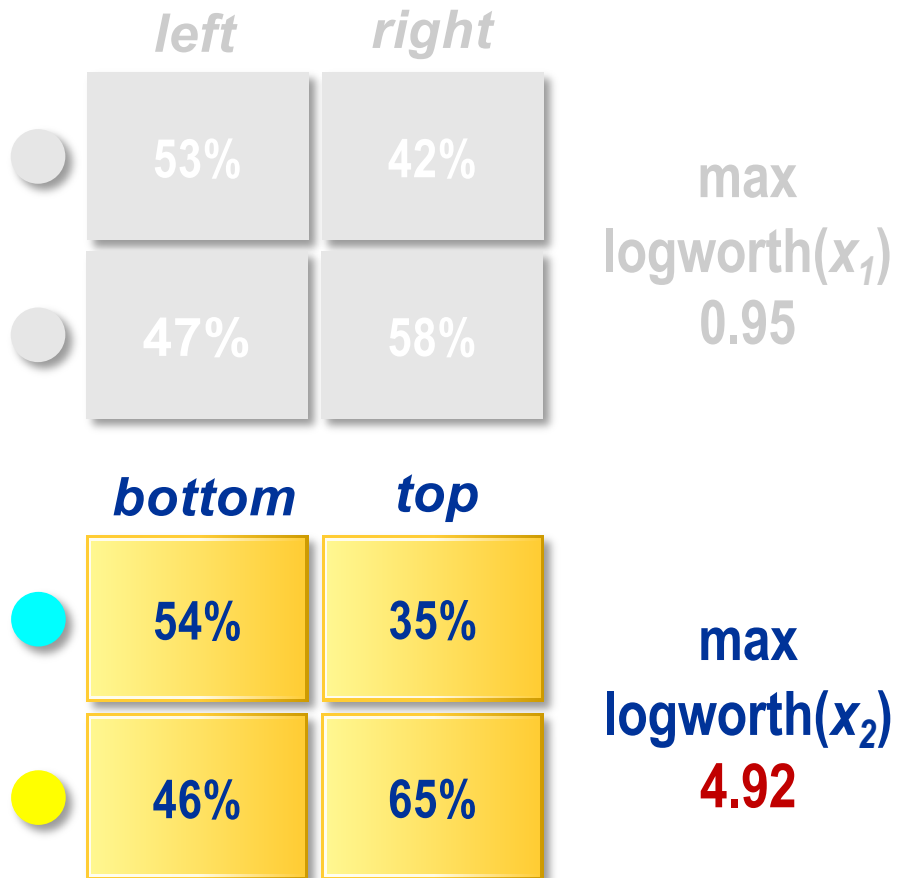
Decision Tree Split Search



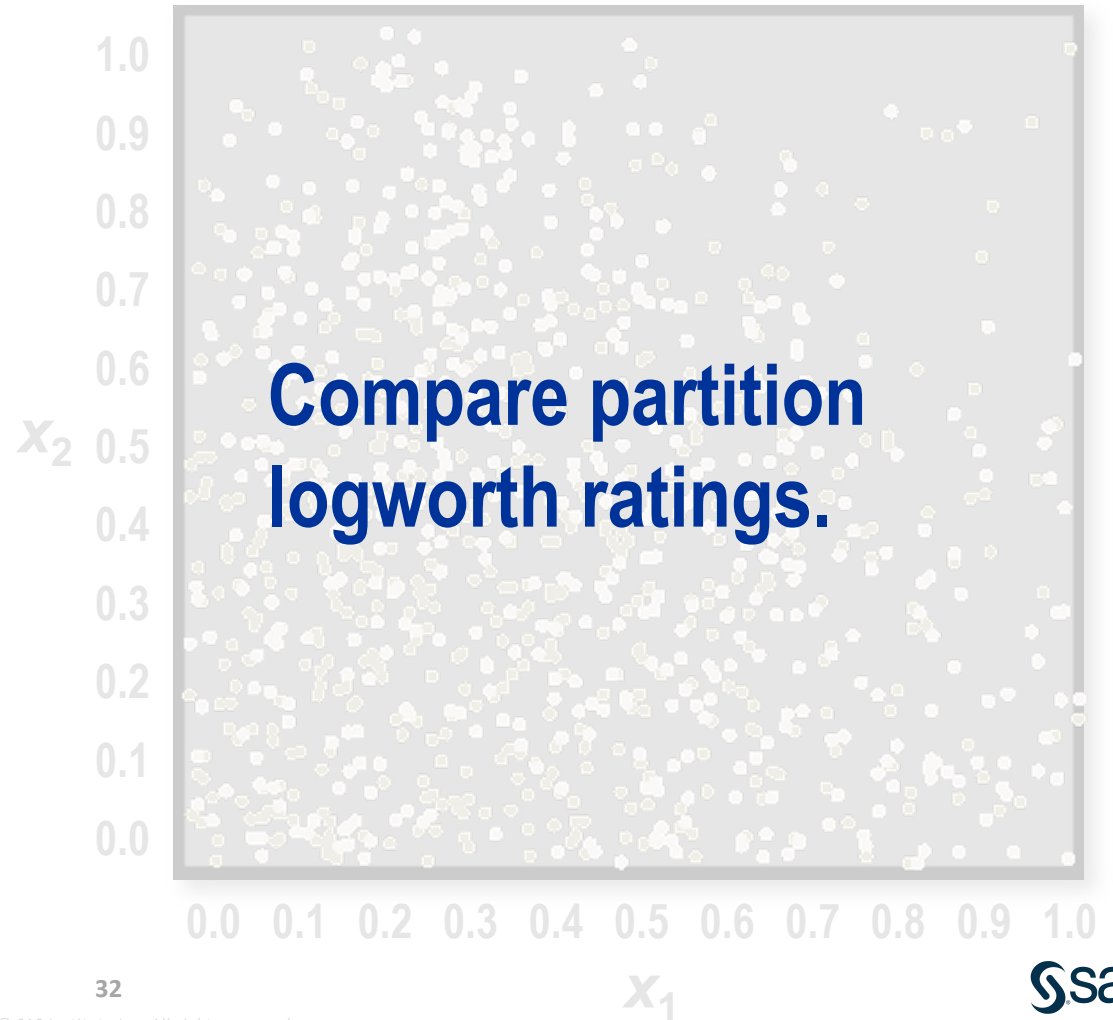
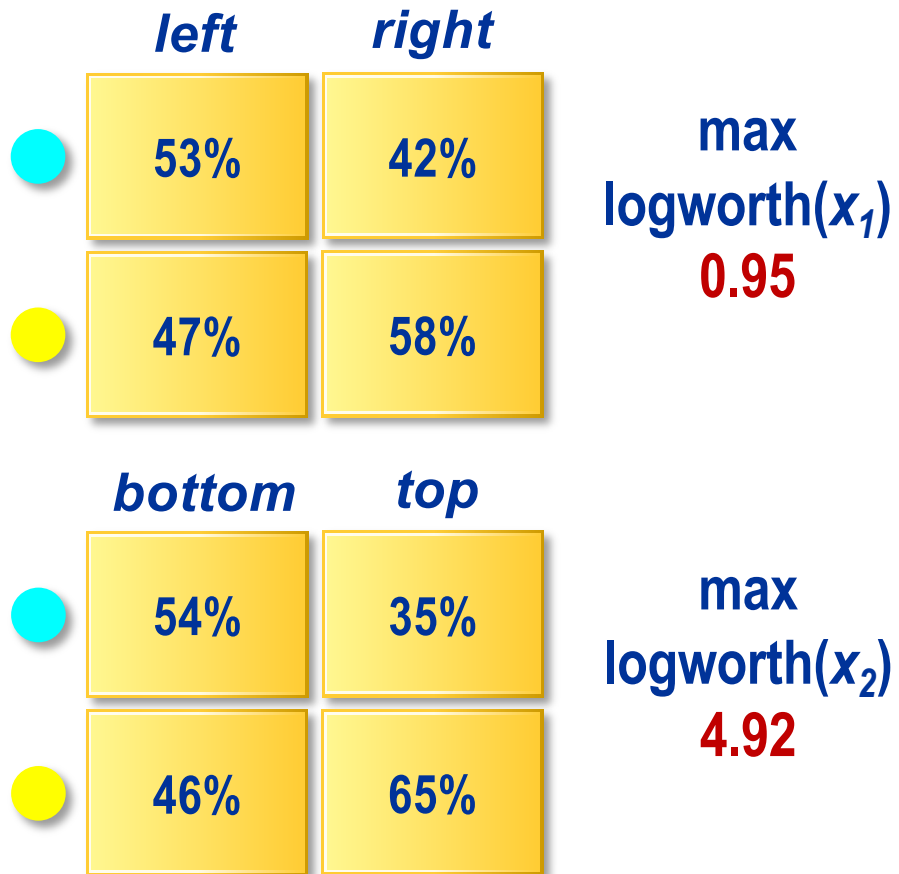
Repeat for input x_2 .



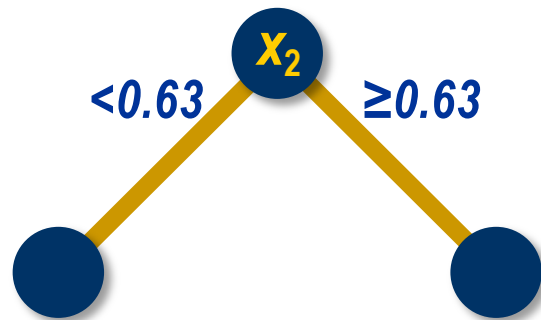
Decision Tree Split Search



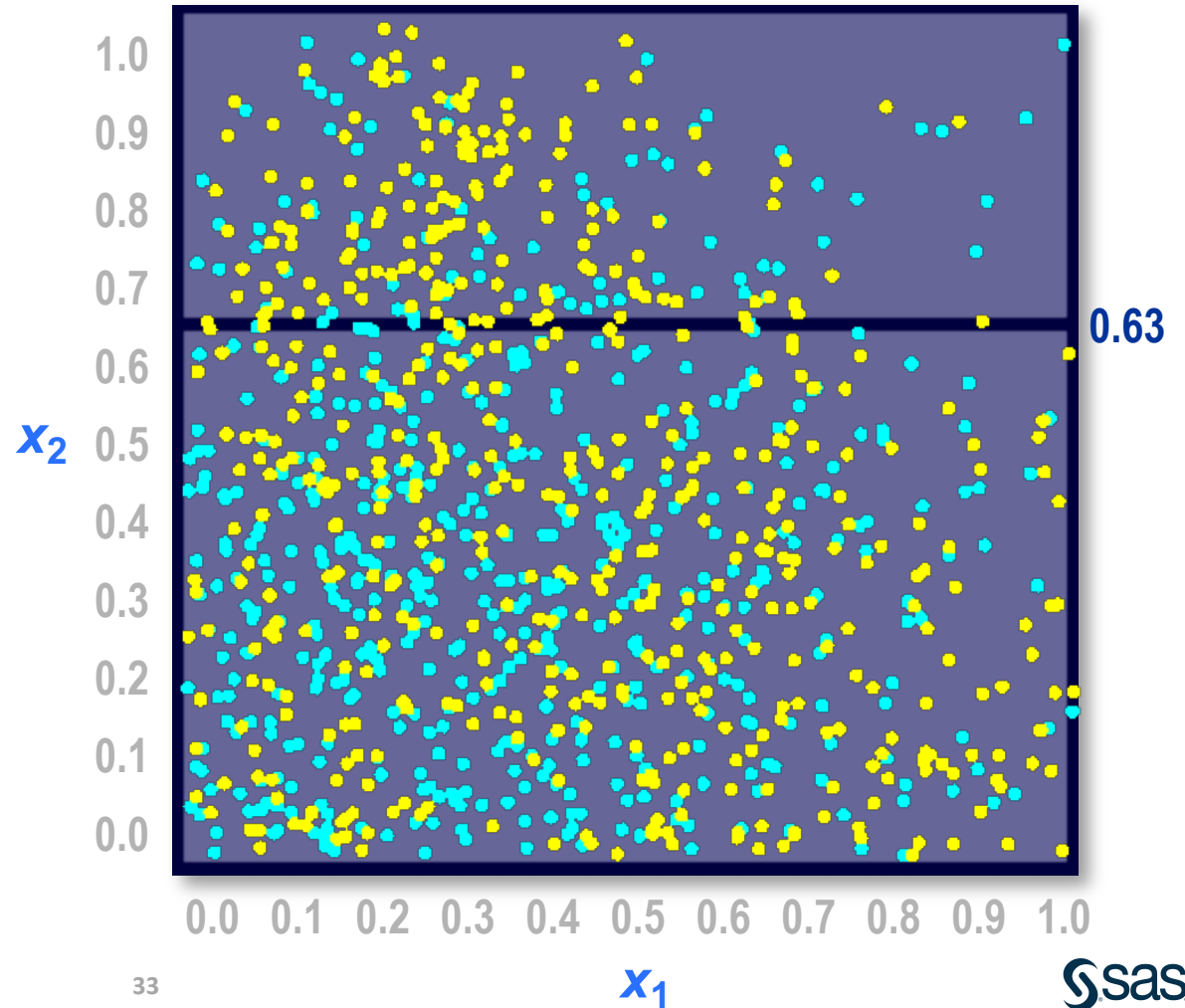
Decision Tree Split Search



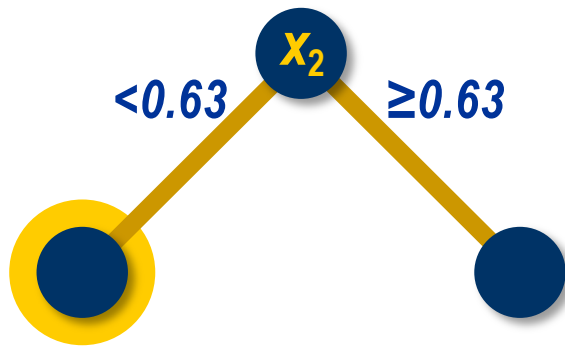
Decision Tree Split Search



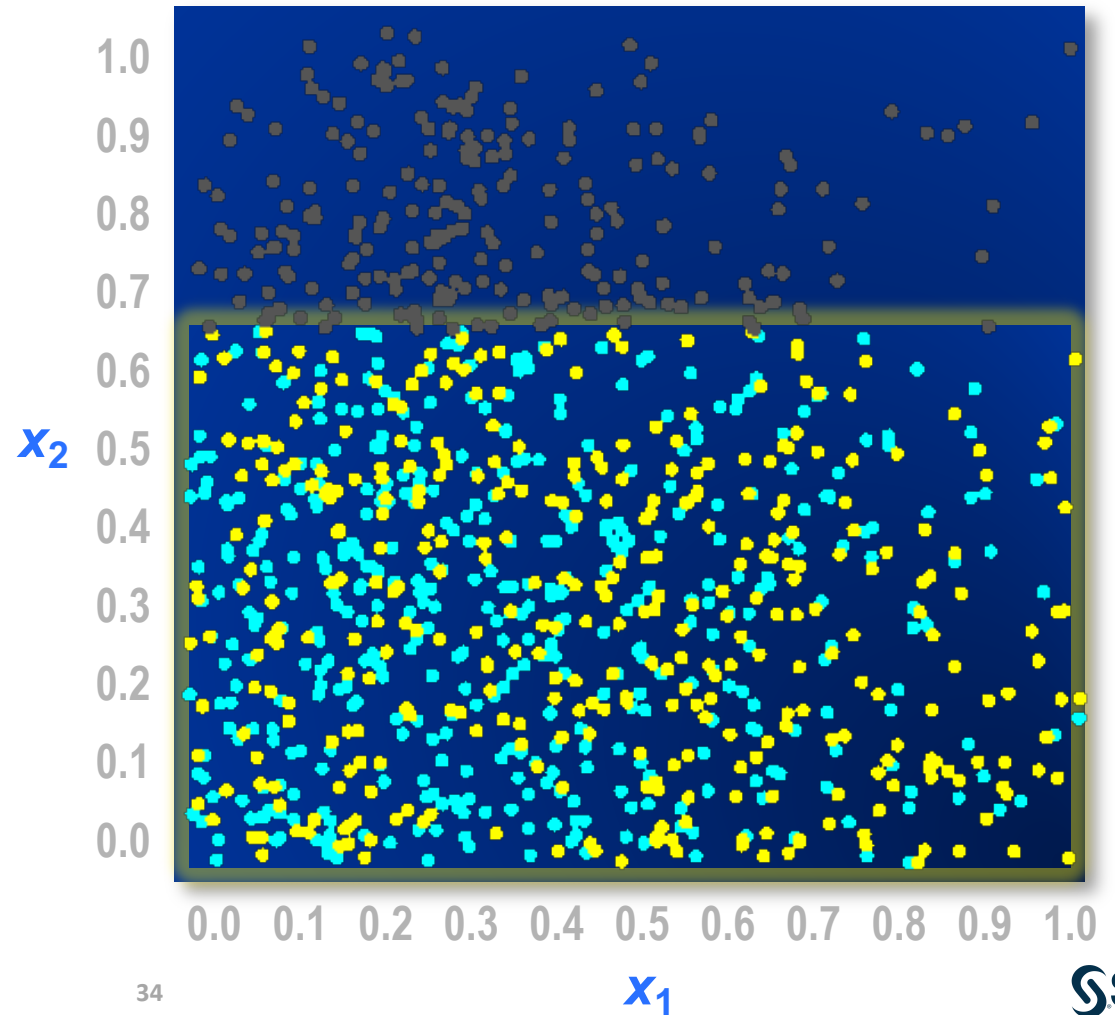
**Create a partition rule
from the best partition
across all inputs.**



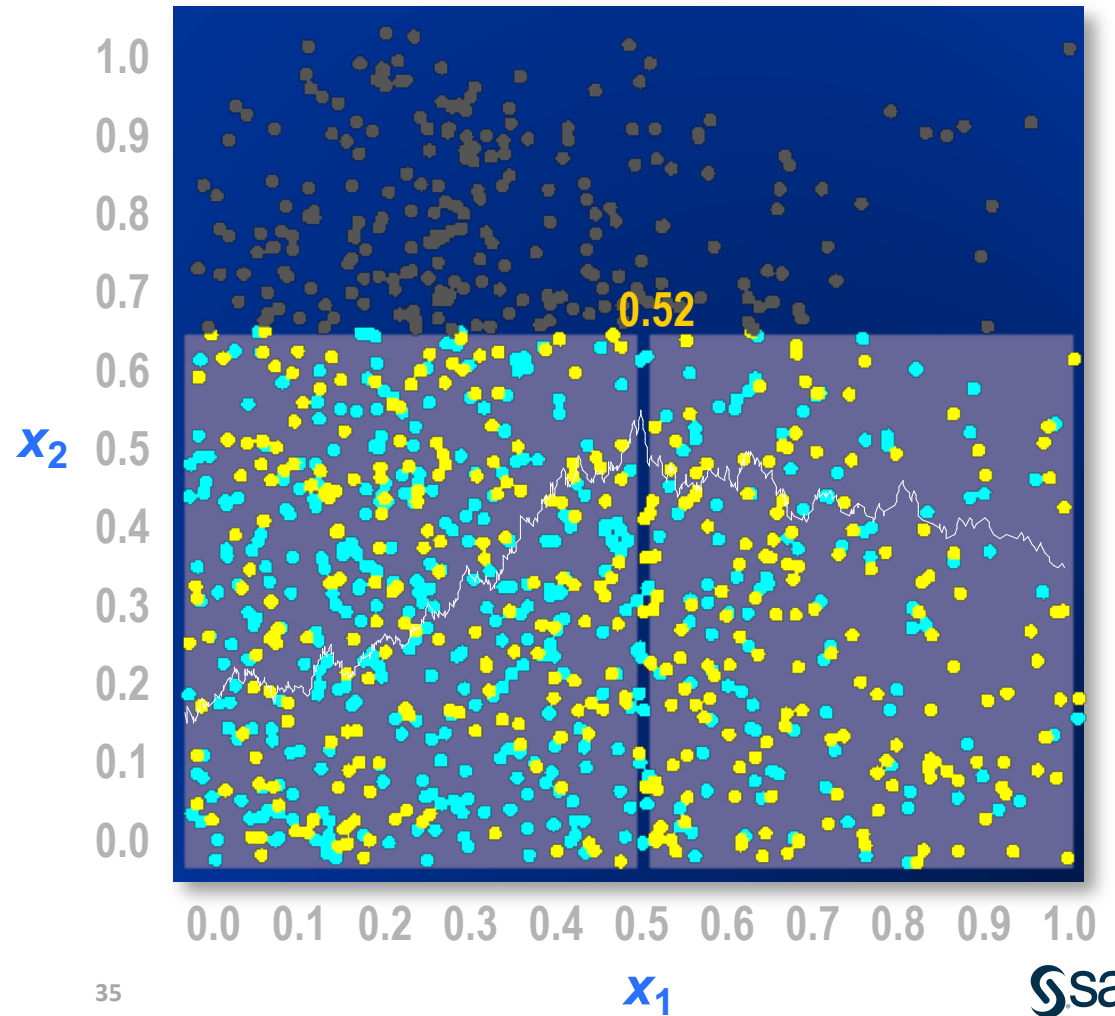
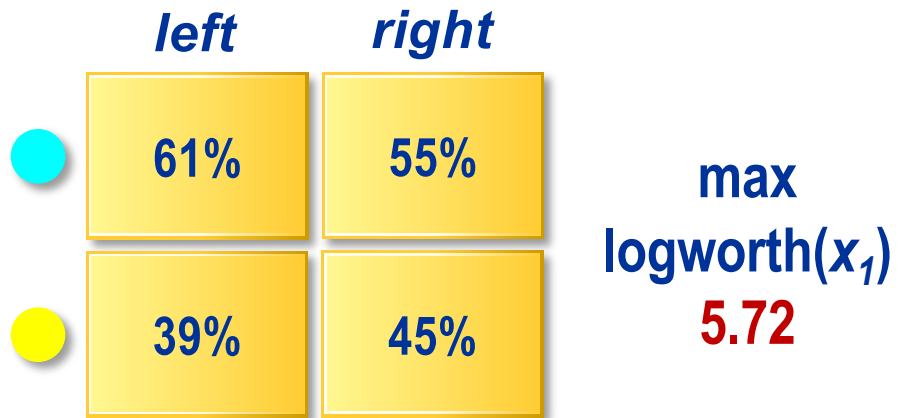
Decision Tree Split Search



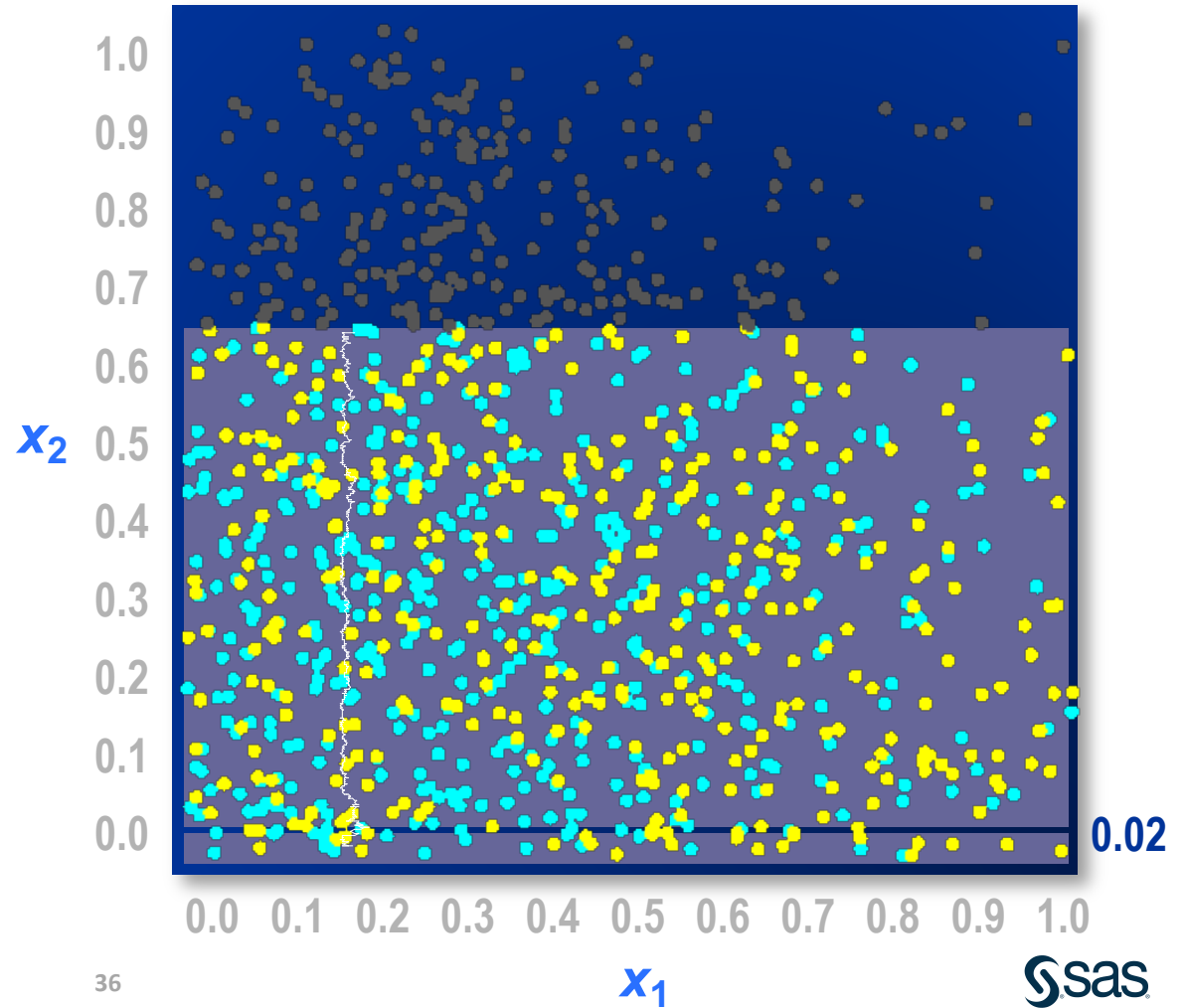
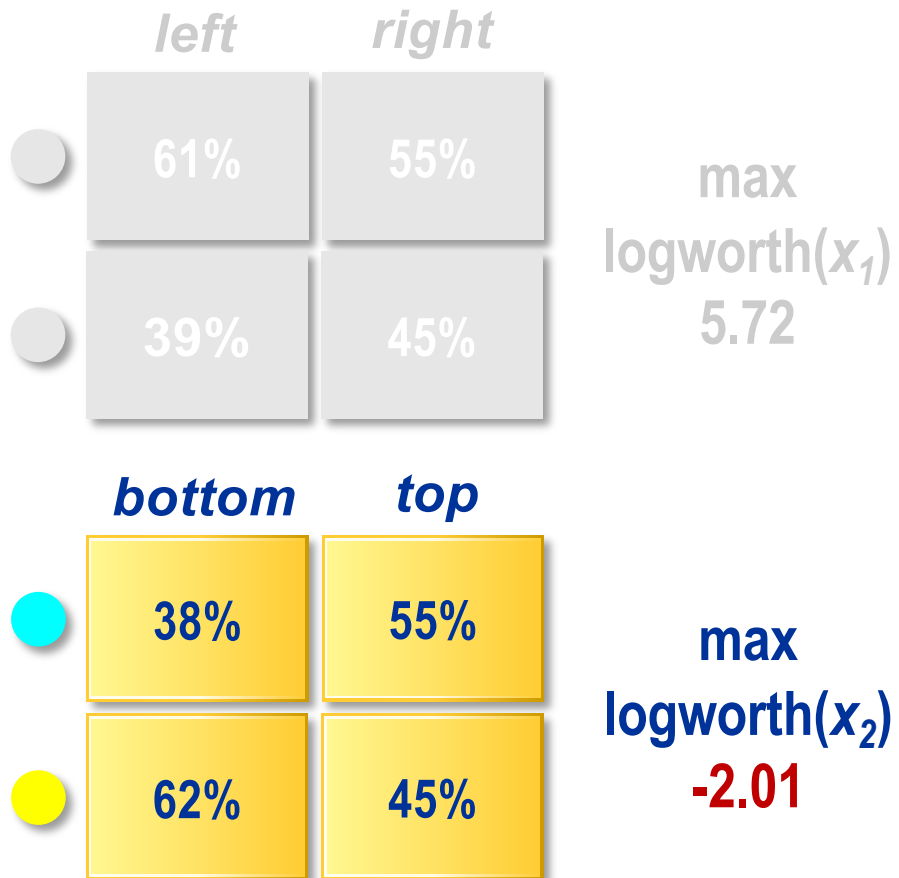
Repeat the process
in each subset.



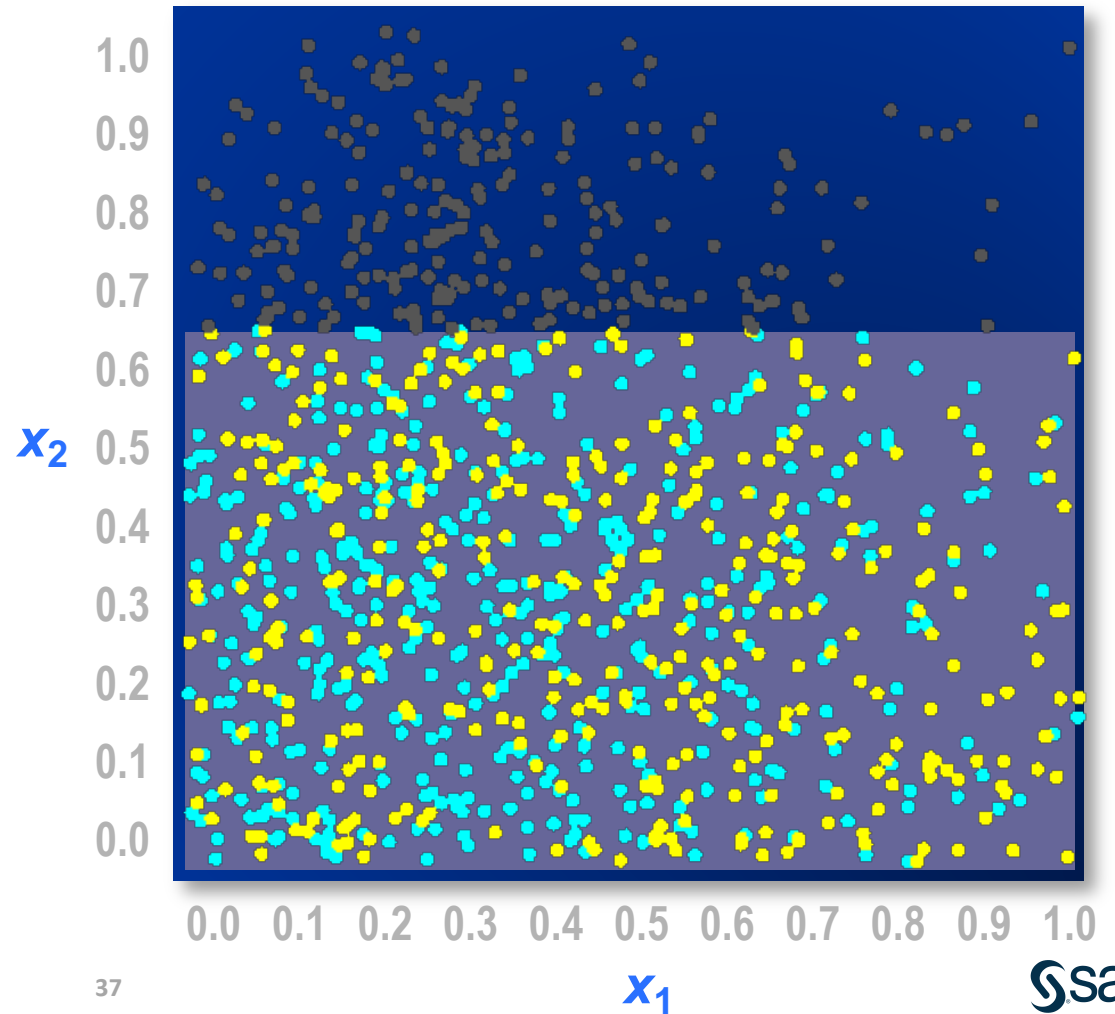
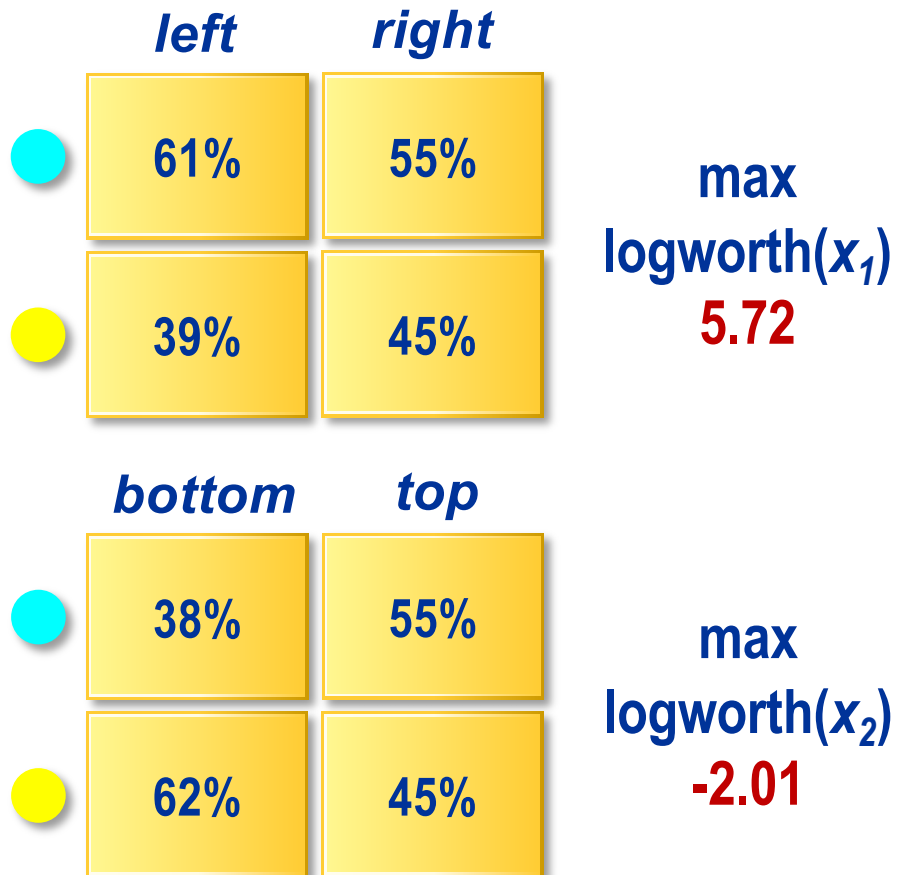
Decision Tree Split Search



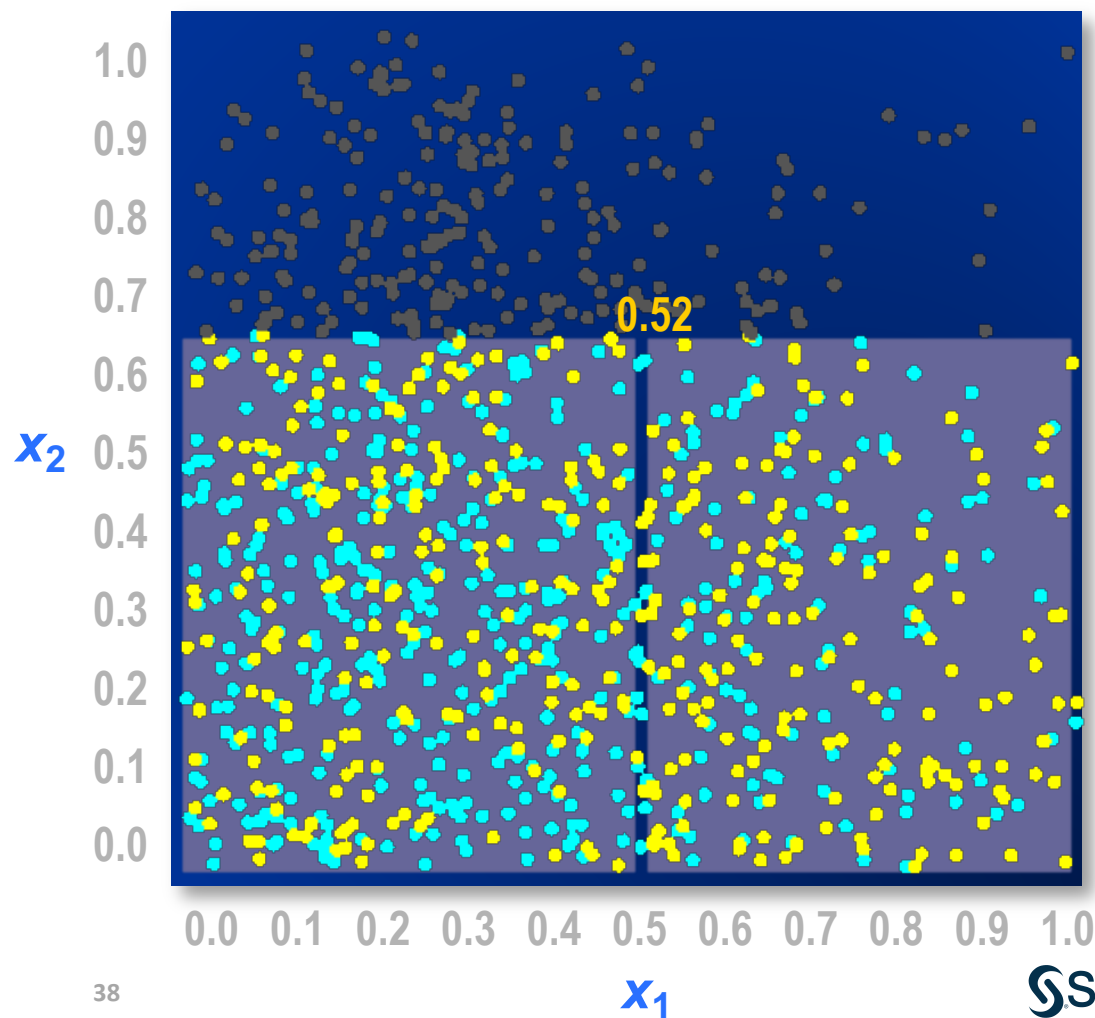
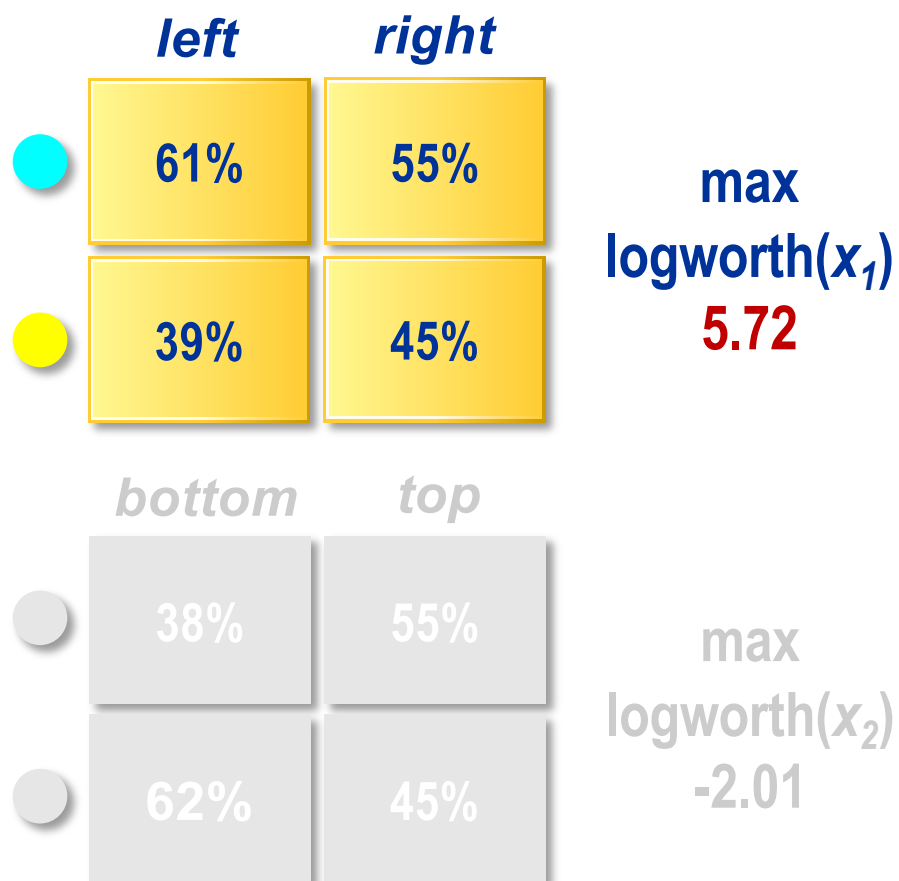
Decision Tree Split Search



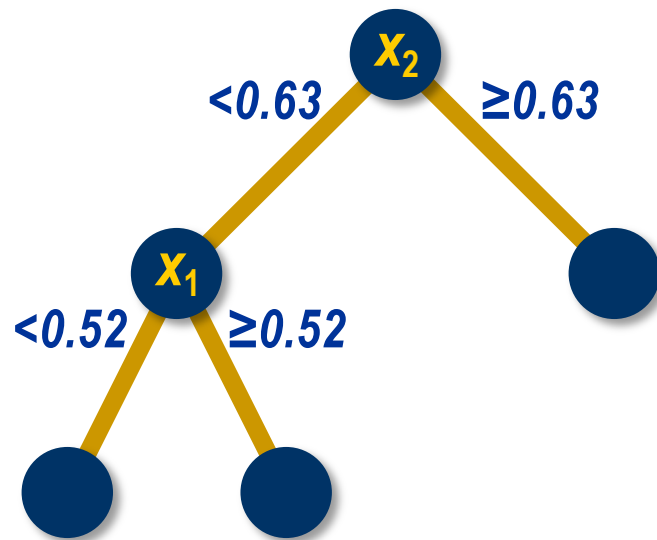
Decision Tree Split Search



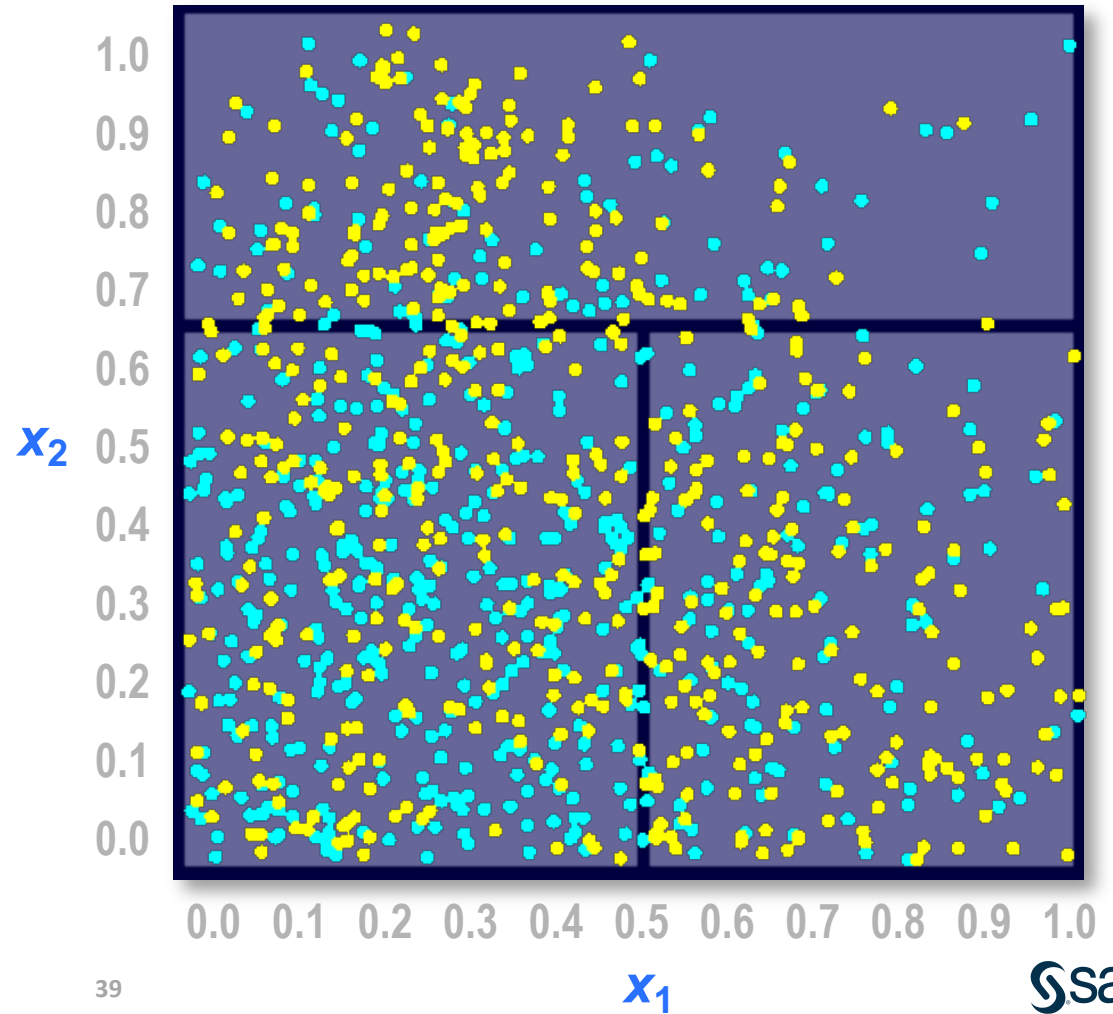
Decision Tree Split Search



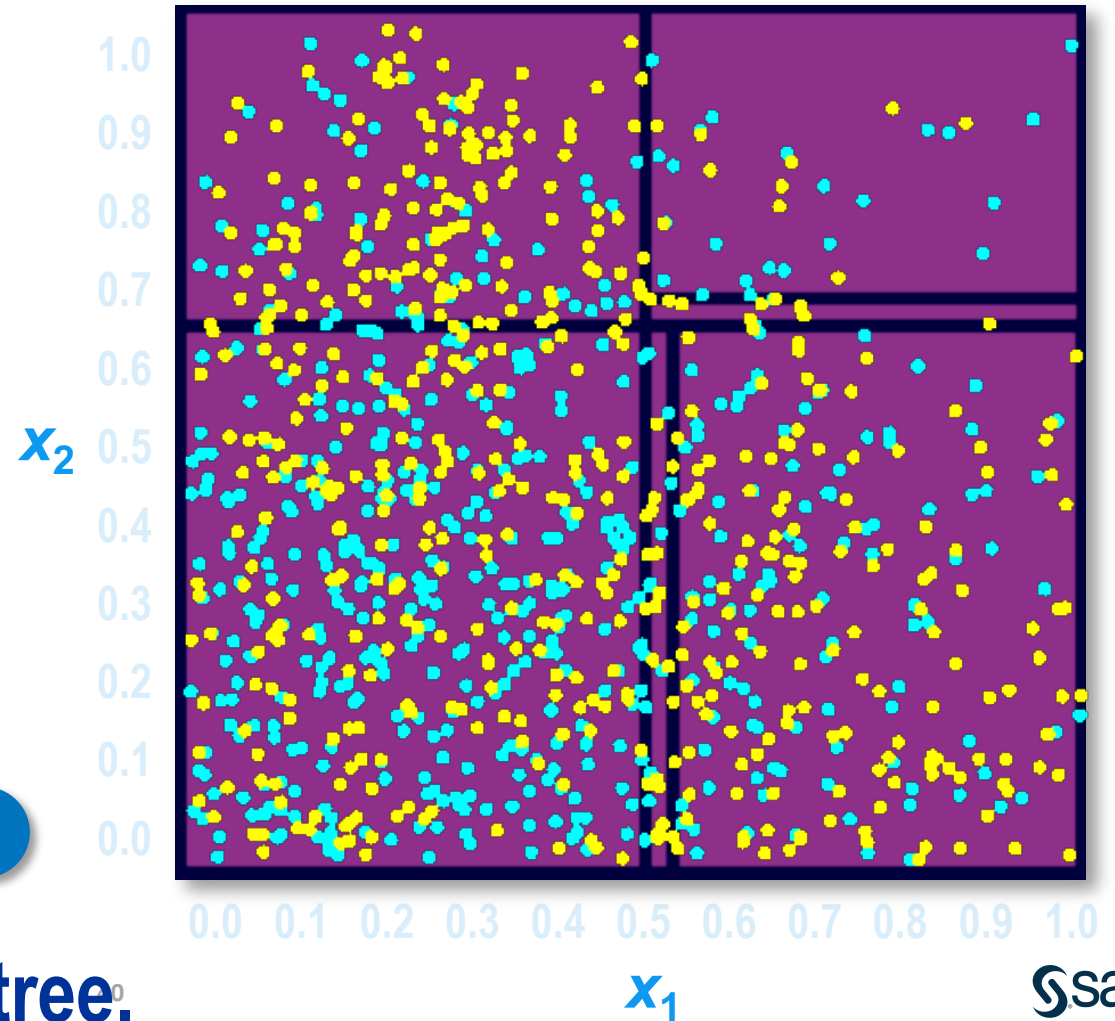
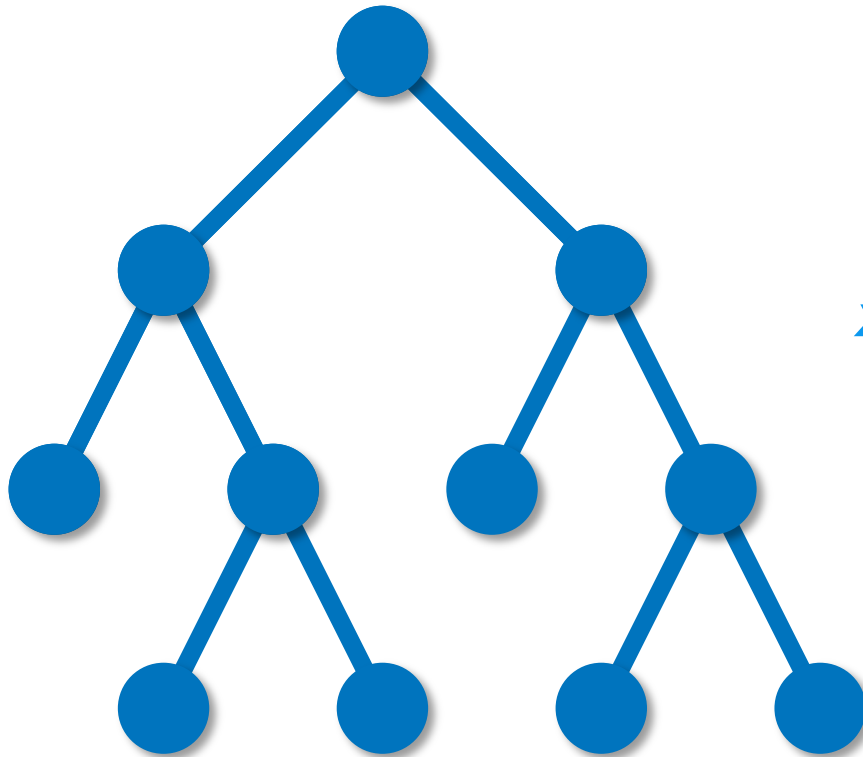
Decision Tree Split Search



Create a second partition rule.



Decision Tree Split Search



Repeat to form a maximal tree.

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Model Essentials: Decision Trees

▶ Predict new cases.

Prediction rules

▶ **Select useful inputs.**

Split search

▶ Optimize complexity.

Pruning



Constructing a Decision Tree Predictive Model

This demonstration illustrates constructing a decision tree model interactively.

Lesson 3: Introduction to Predictive Modeling: Predictive Modeling Fundamentals and Decision Trees

3.1 Introduction

3.2 Cultivating Decision Trees

3.3 Optimizing the Complexity of Decision Trees

3.4 Understanding Additional Diagnostic Tools (Self-Study)

3.5 Autonomous Tree Growth Options (Self-Study)

Model Essentials: Decision Trees

▶ Predict new cases. Prediction rules

▶ Select useful inputs. Split search

▶ **Optimize complexity.** Pruning

Predictive Model Sequence

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



Create a sequence of models with increasing complexity from the *training* data.

Model Complexity

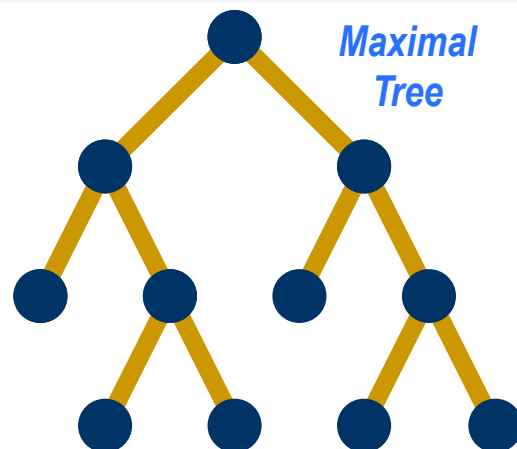
Maximal Tree

Training Data

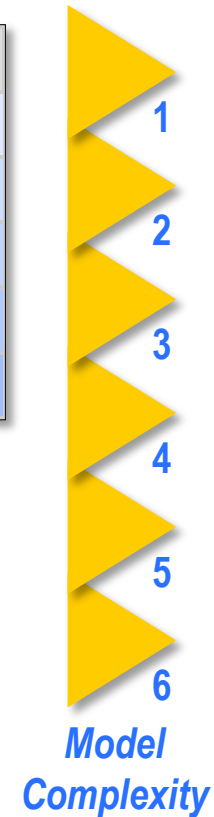
	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



A maximal tree is the most complex model in the sequence.



Create a sequence of models with increasing complexity from the *training data*.

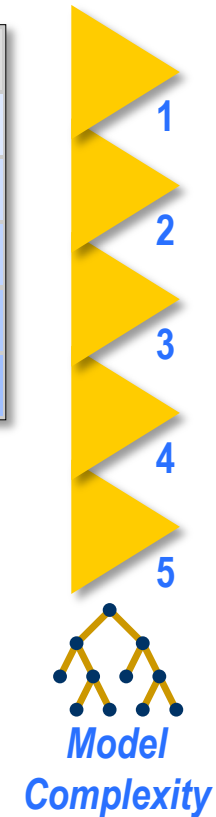
Maximal Tree

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



A maximal tree is the most complex model in the sequence.

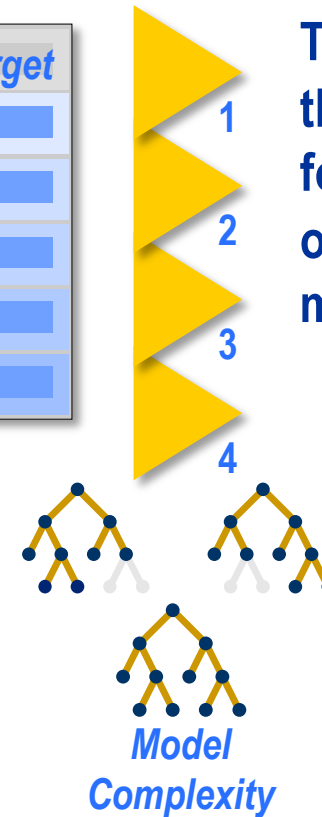
Pruning One Split

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



The next model in the sequence is formed by *pruning* one split from the maximal tree.

Pruning One Split

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



Each subtree's predictive performance is rated on *validation* data.

Pruning One Split

Training Data

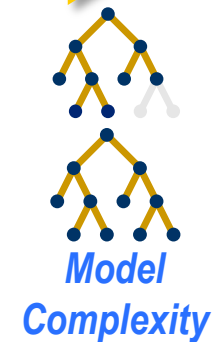
	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



The subtree with the highest *validation* assessment is selected.



Pruning Two Splits

Training Data

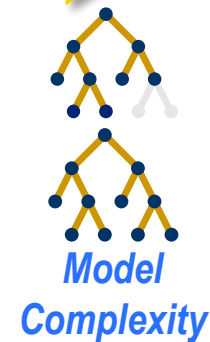
	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



Similarly, this is done for subsequent models.



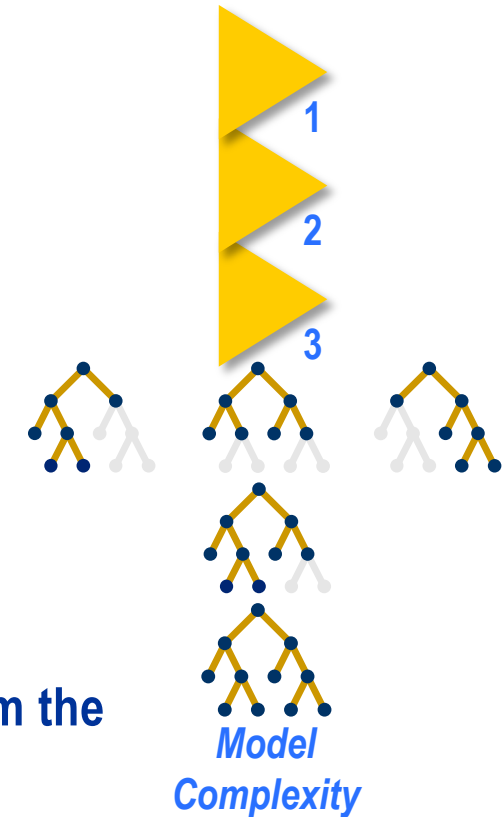
Pruning Two Splits

Training Data

	inputs			target

Validation Data

	inputs			target



Prune two splits from the maximal tree,...

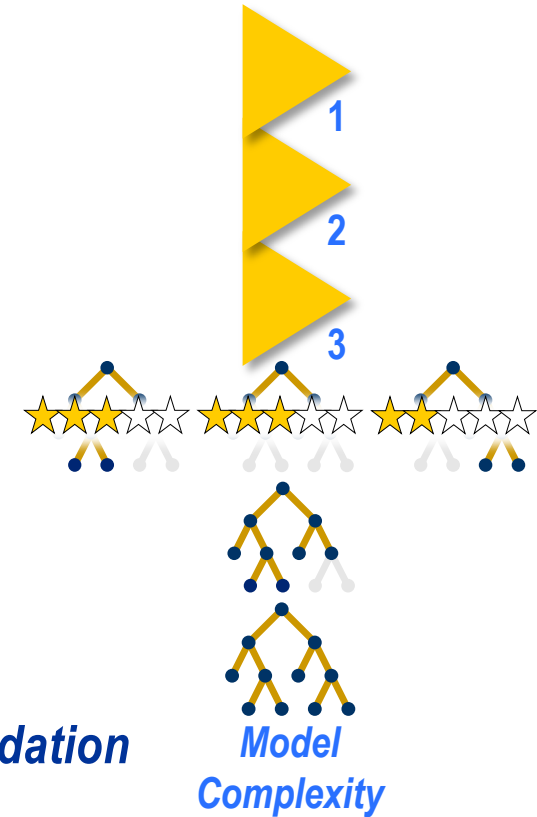
Pruning Two Splits

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



...rate each subtree using *validation* assessment, and...

Pruning Two Splits

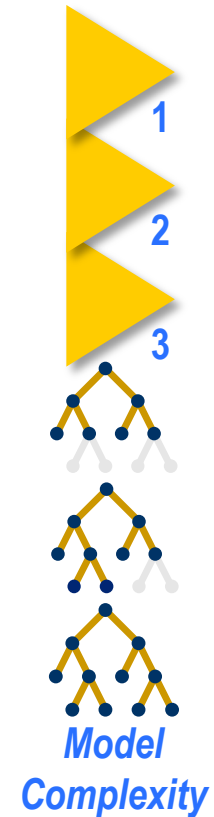
Training Data

	inputs			target

Validation Data

	inputs			target

...select the subtree with
the best assessment rating.



Subsequent Pruning

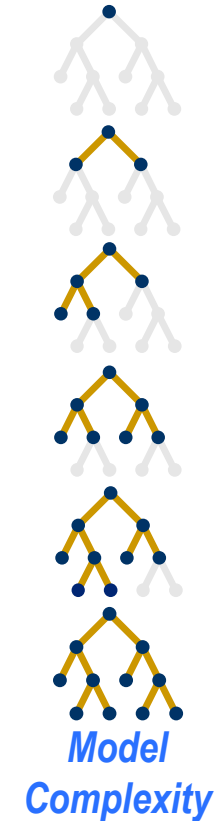
Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

Continue pruning until all
subtrees are considered.



Selecting the Best Tree

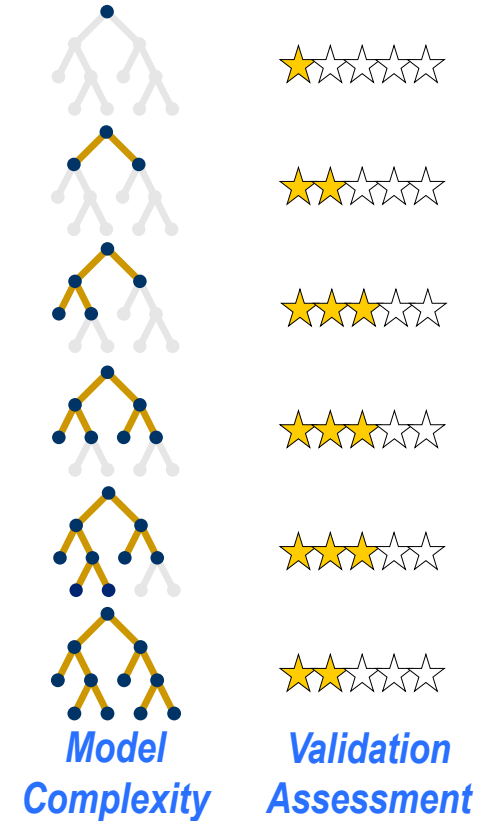
Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

Compare *validation* assessment
between tree complexities.



Validation Assessment

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>

**Choose the simplest model with
the highest *validation* assessment.**



Validation Assessment

Training Data

	<i>inputs</i>			<i>target</i>

Validation Data

	<i>inputs</i>			<i>target</i>



**What are appropriate
validation assessment
ratings?**

Assessment Statistics

Validation Data

	<i>inputs</i>			<i>target</i>

Ratings depend on...



target measurement
(binary, continuous,
and so on)



**prediction type (decisions,
rankings, estimates)**